

**MINISTRY OF EDUCATION AND TRAINING
FPT UNIVERSITY**

**A NEW IMPROVEMENT FOR PREDICT LOAN DEFAULT
USING MACHINE LEARNING**

by

Ngo Tien Luu

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

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Supervisor:

1. Dr. Phan Duy Hung

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Degree Master of Software Engineering

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Preface

In recent years, artificial intelligence, more specifically machine learning, is emerging as evidence of the fourth industrial revolution (1-steam engine, 2-electric power, 3- Information Technology). Artificial intelligence based on big data has become a widespread trend in the world. It has proved to be a great success in theoretical research, experimental, practical application, and is gradually going into all human living areas. Artificial intelligence has been a long development and many significant achievements such as automation cars from Google and Tesla, the face recognition system of Facebook, the virtual assistant Siri of Apple, Cortana of Microsoft, Amazon's product suggestion system, Netflix film suggestion system, Google DeepMind's AlphaGo. In Vietnam, products of pioneering research company groups such as FPT.AI, Viettel.AI, Vin.AI, VNPT.AI, and many other examples are just some of the countless practical applications that artificial intelligence or machine learning gives people.

This thesis builds with the primary purpose of reinforcing the machine learning knowledge equipped while the author studies at the Master of Software Engineering at FPT University. Moreover, the author also tries to apply the learned knowledge into real life to solve one of many problems of risk management at financial and banking institutions are facing. That is the problem of using the machine learning method to predict the probability of loan default.

Although the author has invested a lot of time and effort, the author has limited knowledge and a massive amount of learning. It is these things that lead to unavoidable shortcomings or wrong results. Therefore, the author hopes to receive many comments and feedback to make the essay more complete.

Abstract

In the lending industry, such as banks or finance companies, investors provide loans to borrowers in exchange for the promise of repayment with interest. If the borrower repays the loan, then the lender would make a profit from the interest. However, if the borrower fails to repay the loan, then the lender loses money. Therefore, lenders face predicting the risk of a borrower being unable to repay a loan at any step of the credit process. Many financial institutes calculate FICO scores of a customer based on some factors such as the Payment History (35%), Amounts owed (30%), Length of credit history (15%), Credit mix (10%), and New credit (10% of the total point) of the customer, that score qualifies how risk customers are, how repayment ability of the customer. FICO score is a traditional approach used broadly in finance fields to estimate risks and give a worthy credit amount to the customer. In this study, a new way is to use Machine Learning to predict the above problems. It bases on credit information and more personal information of the customers to predict its repayment ability. The data collected from VietCredit Finances used for training several Machine Learning models to determine if the borrower can repay its loan. Some linear, nonlinear models involved are Logistic Regression, Linear Discriminant Analysis, K Nearest Neighbors, Decision Tree, Naïve Bayes, Support Vector Machine. The algorithm turning and Boosting and Bagging ensemble methods were also used to determine the best model. As a result, the Gradient Boosting model was the outperform and optimal predictive model (PR-AUC metric = 0.95).

Keywords: machine learning, micro-finance, risk management, credit risk, credit score, credit loan assessment, loan repayment ability, probability loan default.

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We want to send the deepest thanks to our teacher Dr. Phan Duy Hung, who has always accompanied the MSE5 class throughout the course. He always has dedicated guidance, sharing experiences, motivating, encouraging, transmitting the flame of a passion to all students.

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List of Abbreviations

Abbreviation	Description
AdaBoost	Adaptive Boosting
ANN	Artificial Neural Network
AUC	Area Under Curve
Bagging	Bootstrap Aggregating
CV	Cross-Validation
LDA	Linear Discriminate Analysis
PCA	Principal Component Analysis
SVM	Support Vector Machine
ROC	Receiver Operating Characteristic
PR	Precision Recall
CART	Classification and Regression Trees
KNN	K-Nearest Neighbors
EML	Extreme Learning Machines

Table 1: Abbreviation.

Chapter 1

Introduction

1.1 Motivation

The size of Vietnam's consumer finance market has continued to have stable growth in recent years, total consumer credit loans estimated from 646,000 billion VND in 2016 will reach one million VND billion (~ US \$ 40.3 billion) in 2019, with an average growth of approximately 29% per year (according to State Bank of Vietnam data) [1]. Along with the development of telecommunications infrastructure, internet, shopping habits on digital platforms of consumers, plus more than half of the population of Vietnam has never used credit through banks, so the amount of consumer finance market is still huge for companies, financial institutions to exploit. From an economic perspective, finance companies play an essential role in the transition of an export-dependent economy to one that depends on domestic consumption. In terms of social aspects, finance companies meet the needs of people to borrow money for consumption, mainly low and middle-income people, who have difficulty accessing bank loans. It helps them have the opportunity to borrow money from a more reliable address, thereby reducing black credit activities, contributing to economic development, and improved quality of life [2], [3].

The consumed loan is one significant product of any financial institutes. All financial companies are trying to figure out effective business strategies to persuade more customers to apply for their loans [8]. With simple loan procedures, no need for collateral, an easy approval process, and quick disbursement cause the number of consumed loan applications at financial companies increases, taking a lot of time and resources to process promptly [4], [8]. To minimize credit risks leading to bad debts, ensure financial safe for customers and organizations, loan appraisal is always the most critical stage in the credit approval process [6], [10]. Therefore, many financial institutions take several input variables (such as personal information, monthly income, expenses, credit history) into account before approving a loan and determining whether a given borrower will fully pay off the loan or otherwise [10], [11], [12], [19]. If the lender is too strict, fewer loans get approved, which means there is less interest to collect. However, if they are too lax, they end up approving loans that default.

1.2 Literature Review

This part of the thesis is studying different research papers comparing the classification techniques for loan default prediction. Some positive results are achieved by many researchers who conducted studies on the loan default prediction.

In [4], R. E. Turkson and et al. examined actual bank credit data, conducted several machine learning algorithms on the data for comparative analysis, and chose which algorithms best fit learning bank credit data. The experiment revealed that, apart from the Nearest Centroid and Gaussian Naive Bayes, the rest of the algorithms perform credibly well in terms of their accuracy and other performance evaluation metrics. Each of these algorithms achieved an accuracy rate between 76% to over 80%. Furthermore, the essential features determine whether a customer will default or otherwise pay credit the next month. They then applied these essential features on some selected machine learning algorithms and compared their predictive accuracy with the other algorithms that used all the 23 features. The results show no significant difference, signifying that these features can accurately determine the customers' creditworthiness. Finally, the authors formulate a predictive model using the most important features to predict the creditworthiness of a given customer.

Zhu, Lin Qiu, Dafeng Ergu, Daji Ying, Cai Liu & Kuiyi (2019) in research [5] used the Random Forest algorithm to build a loan default prediction model given the real-world user loan data on Lending Club. The SMOTE method is adopted to cope with the problem of imbalanced class in the dataset, and then a series of operations such as data cleaning and dimensionality reduction is carried out. The experimental results show that: Random Forest algorithm outperforms compared with the other three machine learning methods, namely the decision tree, the logistic regression, and the SVM. The ROC-AUC of Random Forest reached 0.983.

Sheikh and et al. used Logistic Regression, SVN, KNN models to predict the repayment capability of credit loans. Data sets used to train and test the model were collected on Kaggle. On the Data set, the best case accuracy obtained on the original data set is 0.811. The following conclusions are reached after analysis that those applicants whose credit score was worst will fail to get loan approval due to a higher probability of not paying back the loan amount. Most of the time, those applicants who have high income and demand a lower amount of loan are more likely to get

approved, which makes sense, more likely to pay back their loans. Some other characteristics like gender and marital status seem not to be considered by the company [6].

L. Soares de Melo Junior and et al. evaluated several classification approaches to credit scoring. Besides, They also assessed some pre-processing methods to overcome skewed datasets. To achieve it, they used three public real-world credit scoring datasets. Their experiments progressively increase the class imbalance in each of these datasets by randomly undersampling the minority class of defaulters to identify how the predictive power is affected. The results indicate that Random Forest, Extreme Gradient Boosting perform very well in all imbalance levels. They also found that a complete grid search step can increase the prediction power of classification approaches in high imbalanced datasets [7].

Aslam UAziz HSohail A et al., in their study [8], reviewed machine learning techniques such as Decision trees, Neural Networks, Logistic Regression, and Support Vector Machine. Each machine learning algorithm proved to be accurate and valuable in different settings. Furthermore, although most studies focused on accuracy for the prediction of loan default detection, it was noticed that few studies focused on the repercussions of false negatives, which can be pretty devastating to the lending companies. Therefore, it is advised to future researchers to focus more on false negatives when approaching the loan lending prediction problem.

In research [9], S. Z. H. Shoumo and et al. proposed a machine learning model which can precisely assess credit risk and predict possible loan defaulters for credit lending institutions. A comparative analysis has made using tuned supervised learning algorithms such as Support Vector Machine, Random Forest, Extreme Gradient Boosting, and Logistic Regression to identify defaulters. Recursive Feature Elimination with Cross-Validation and Principal Component Analysis has been used for dimensionality reduction. Metrics such as F1 score, AUC score, prediction accuracy, precision, and recall have been used to evaluate each model. Among all the models, the combination of a tuned Support Vector Machine and Recursive Feature Elimination with Cross-Validation has shown great promise in identifying loan defaulters. Therefore, the proposed model can assist financial institutions in accurately identifying loan defaulters and preventing them from incurring further losses.

T. N. Pandey, A. K. Jagadev, S. K. Mohapatra, and S. Dehuri surveyed the different classifiers used in the credit risk evaluation. In this paper, different classifiers are discussed, and different

types of ensemble classifiers are briefed. They found that the ELM classifier gives better accuracies compare to other classifiers that are ELM gives 96.33(%) in the German dataset and 96.32(%) in the Australian dataset [10].

S. J. Shiv and et al. in research [11] studied the performance of various classification algorithms applied to a real-life data set for consumer lending. Various classifiers (logistic regression, random forest, neural networks) were evaluated, and their performance was compared. The data set first pre-processed to select the best attributes and to remove outliers for classification. The performance assessed using various metrics, including classification accuracy, precision, recall, and specificity. It found that Logistic Regression yields excellent performance.

In the personal study, W. Qiu reduced the negative impact of imbalanced data set on the model, and he constructed a credit rating model based on K-XGBoost from the perspective of a hybrid model, which is a new hybrid model proposed by this paper combining the advantages of both KMeans++ and XGBoost. Moreover, he constructs XGBoost credit rating models based on SMOTE series algorithms from the perspective of random oversampling. According to empirical analysis, the indicators of the K-XGBoost model are superior to the XGBoost model of dataset balanced by the Borderline SMOTE algorithm and can reduce the unnecessary costs by accelerating modeling. To meet the needs of commercial banks for risk control and market share, the author constructed a three-class credit rating model based on XGBoost, which divides customers into good stable customers, unstable good customers, and wrong customers. When the loan policy is tightened, only stable customers will be the target customer group. Empirical research shows that the above three models have good performance and strong generalization ability [12].

Machine learning was adopted to design a prediction process, to solve the credit evaluation problem of online P2P lending. A multi-stage ensemble learning model was proposed to evaluate the borrowers' credits, in which the Gradient Boosting Decision Tree (GBDT) algorithm used for feature mapping, and a special Auto-Encoder applied to extract the best features. Furthermore, the Logical Regression algorithm is designed to classify the borrowers. Through computational experiments on the open data set and taking AUC (Area Under Curve) as evaluation index, the effectiveness of the proposed model and algorithms testified [13].

In group research [14], The Ensemble GradientBoost successfully improves the performance of a single Logistic Regression classification and achieves a higher degree of accuracy across both datasets. The German dataset yields 81% accuracy and 88.4% of the Australian dataset.

L. E. Boiko Ferreira and et al., in their study [15], wrangled a real-world P2P lending data set from Lending Club, containing a large amount of data gathered from 2007 up to 2016. The authors analyzed how supervised classification models and techniques handle class imbalance. Ensembles, cost-sensitive, and sampling methods are combined and evaluated along with Logistic Regression, Decision Tree, and Bayesian learning schemes. Results show that, on average, sampling techniques outperform ensembles and cost-sensitive approaches.

The group authors include Chen, Ya Qi Zhang, Jianjun Ng, Wing W.Y., who focused on dealing with the data imbalance problem to enhance the performance of the loan default prediction. The approach in this study is a hybrid undersampling method that combines the clustering, the stochastic sensitivity measure, and the radial basis function neural networks. Actual loan default data from a P2P company in China used to validate the method's performance. Experiments' results demonstrate that their approach yields a better generalization performance [16].

Barboza and et al. tested machine learning models (Support Vector Machines, Bagging, Boosting, and Random Forest) to predict bankruptcy one year before the event and compare their performance with results from the Discriminant Analysis Logistic Regression and Neural Networks. Authors used data from 1985 to 2013 on North American firms, integrating information from the Salomon Center database and Compustat, analyzing more than 10,000 observations. Comparing the best models with all predictive variables, the machine learning technique related to Random Forest led to 87% accuracy, whereas Logistic Regression and Linear Discriminant Analysis led to 69% and 50% accuracy, respectively, in the testing sample [17].

In [18], this study presents an empirical comparison of various combinations of classifiers and resampling techniques within a novel risk assessment methodology that incorporates imbalanced data. The credit predictions from each combination were evaluated with a G-mean measure to avoid bias towards the majority class, which was not considered in similar studies. The results reveal that combining Random Forest and random under-sampling may be an effective strategy for calculating the credit risk associated with loan applicants in social lending markets.

Coşer, Alexandru, Monica Mihaela Maer-Matei, and Crişan Albu. 2019. The paper title "Predictive Models for Loan Default Risk Assessment." proposes an analysis of a database of customers where a part of them was unable to repay their loans and got into default status. Using the methodology of data mining and machine learning algorithms, a series of predictive models developed using classifiers such as LightGBM, XGBoost, Logistic Regression, and Random Forest to evaluate the probability of a customer's entering loan default. Three sampling scenarios were created to compare the classification between imbalanced and balanced data sets. Moreover, a model comparison analysis was performed to identify the best classifier by considering the model performance metrics: AUC score, Precision, Recall, and Accuracy. The best results observed for the Random Forest optimal classifier applied on the combined scenario under-over sampling, with a representative AUC of 0.89 [19].

No	Authors	Title	Source Dataset	Imbalanced Dataset	Classifiers	The Best Model	Metric	Score
4	R. E. Turkson, E. Y. Baagyere, and G. E. Wenya	A machine learning approach for predicting bank creditworthiness	UCI		ExtraTree, RandomForest, AdaBoost, GradientBoosting, RandomTreesEmbedding, Voting, LogisticRegression, SVM, CART, NearestCentroid, KNeighbors, GaussianNB, Neural networks, LinearDiscriminantAnalysis, Bagging	AdaBoost	F1	0.9
5	Zhu, Lin, Dafeng Qiu, Daji Ergu, Cai Ying, and Kuiyi Liu	A Study on Predicting Loan Default Based on the Random Forest Algorithm	Lending Club	x	Random Forest, Decision Tree, SVM, Logistic Regression	RandomForest	ROC AUC	0.983
6	M. A. Sheikh, A. K. Goel, and T. Kumar	An Approach for Prediction of Loan Approval using Machine Learning Algorithm	Kaggle		logistic regression, SVM, KNN	LogisticRegression	F1	0.811
7	L. Soares de Melo Junior, F. M. Nardini, C. Renso and J. A. Fernandes de Macêdo	An Empirical Comparison of Classification Algorithms for Imbalanced Credit Scoring Datasets	UCI	x	kNN, ELM, SVM, LOGR, TREE, ANN, FUZZ, QDA, GNET, LDA, GNB, RNDF, XGB, ADAB	random forest, extreme gradient boosting	ROC AUC	0.932

N o	Authors	Title	Source Dataset	Imbalanced Dataset	Classifiers	The Best Model	Metric	Score
8	Aslam, Uzair, Hafiz Ilyas Tariq Aziz, Asim Sohail, and Nowshath Kadhar Batcha	An Empirical Study on Loan Default Prediction Models.	UCI		Support Vector Machine, Logistic Regression, Decision Trees, Neural Nets	LR	AUC	0.74
9	S. Z. H. Shoumo, M. I. M. Dhruva, S. Hossain, N. H. Ghani, H. Arif, and S. Islam	Application of Machine Learning in Credit Risk Assessment: A Prelude to Smart Banking	Lending Club	x	LR, SVM, RF, XGB	SVM+RFE	PR AUC	0.99
10	T. N. Pandey, A. K. Jagadev, S. K. Mohapatra and S. Dehuri	Credit risk analysis using machine learning classifiers	UCI		Bayesian classifier, Naive-Bayes, Decision tree, KNN, K-means, MLP, SVM, ANN	ELM	Accuracy	0.96
11	S. J. Shiv, S. Murthy and K. Challuru	Credit Risk Analysis Using Machine Learning Techniques	UCI		LR, RF, NN	LR	Accuracy	0.81
12	W. Qiu	Credit Risk Prediction in an Imbalanced Social Lending Environment Based on XGBoost	Kaggle	x	KXGB, XGB, LR, RF, GBDT	XGB	Accuracy	0.92

N o	Authors	Title	Source Dataset	Imbalanced Dataset	Classifiers	The Best Model	Metric	Score
13	S. Chen, Q. Wang and S. Liu	Credit Risk Prediction in Peer-to-Peer Lending with Ensemble Learning Framework	Kaggle		GBDT_AE_LR, Logistic Regression, Decision Tree, GBDT_LR	Gradient Boosting Decision Tree (GBDT)	ROC AUC	0.858
14	A. Lawi, F. Aziz and S. Syarif	Ensemble GradientBoost for increasing classification accuracy of credit scoring	UCI		Logistic Regression, Bagging, GradientBoost	GradientBoost	Accurac y	0.884
15	L. E. Boiko Ferreira, J. P. Barddal, H. M. Gomes and F. Enembrec k	Improving Credit Risk Prediction in Online Peer- to-Peer (P2P) Lending Using Imbalanced Learning Techniques	Lending x Club		DT, GNB, LG, AdaBoost, Bagging, RF,	LG+RU	ROC AUC	0.66
16	Y. Chen, J. Zhang and W. W. Y. Ng	Loan Default Prediction Using Diversified Sensitivity Undersamplin g	Lending x Club		ROS+LR, ROS+RF, ROS+XGB, RUS+ LR, RUS+ RF, RUS+ XGB, SMOTE+LR, SMOTE+RF, SMOTE+ XGB, DSUS	DSUS	PR	0.688 4
17	Barboza, Flavio, Herbert Kimura, and Edward Altman	Machine Learning Models and Bankruptcy Prediction	North America n		LDA, LR, NN, SVM, Boosting, Bagging, RF	Bagging	ROC AUC	0.914

N o	Authors	Title	Source Dataset	Imbalanced Dataset	Classifiers	The Best Model	Metric	Score
18	Namvar, Anahita, Mohammad Siami, Fethi Rabhi, and Mohsen Naderpour	Credit Risk Prediction in an Imbalanced Social Lending Environment	Lending Club	x	RF-RUS, LDA-SMOTE, LR-Tomek, Logistic Regression, Random forest	LR-RUS	ROC AUC	0.703
19	Coşer, Alexandru, Monica Mihaela Maer-Matei, and Crişan Albu	Predictive Models for Loan Default Risk Assessment.		x	LightGBM, XGBoost, RF, LR	Random Forest optimal	ROC AUC	0.89

Table 2: Literature Review Resume

It is no doubt to say that most of the studies mentioned above aim to try various machine learning techniques to predict loan default and credit risk. Many supervised algorithms are tested, such as linear: Logistic Regression, Linear Discriminant Analysis, Nonlinearity: KNN, Naïve Bayes, Decision Tree, SVM, combined methods are also tested such as Bagging, Boosting, Voting. The data using model training was mainly collected from sharing communities such as UCI, Kaggle, Lending Club. Some authors mention the data imbalance problem as a challenge in building a classification model. The results print out most of the studies have found good models on the data sets collected. However, some studies on imbalanced data using Accuracy or ROC AUC metric lead to excessive optimism in the evaluation results. On the whole, all machine learning methods for this problem are heavily dependent on the data, the features extracted from the data, and the data preparation process.

This study uses several machine learning models to analyze the loan approval process. The dataset in this paper collects from VietCredit Finance, a consumer finance company in Vietnam. It includes approximately 48000 loan applications with 68 features disbursement through the VietCredit between May 2020 and August 2020. The data labeled with 0 (good 73 %) and 1 (bad 27%). This work exploratories data to get more knowledge about the dataset such as data type proportion, input-output variables distribution, check and handle the missing values, outlier detection, correlation input variables with another, between input with output variable, remove redundant variables, and necessary data transformations conducted to process the data. After, several suitable machine

learning models trained on that data set to predict the loan repayment ability of the customer. The linear and nonlinear machine learning models include Logistic Regression, Linear Discriminant Analysis, K Nearest Neighbors, Decision Tree, Naïve Bayes, Support Vector Machine. The Boosting (AdaBoost, Gradient Boosting) and Bagging (Random Forests, Extra Trees) ensemble methods are also used for evaluation. Each model will be evaluated via k-fold classification techniques and using the Precision and Recall Curve (AUC) metric. At the evaluation step with the default parameter set, most classification models work well with a PR-AUC score greater than 0.8 (baseline score is 0.5). However, the LGBMClassifier (Gradient Boosting) has the best performance with a PR-AUC score = 0.95, so the LGBMClassifier is selected as the best model for predicting the held-out data set.

1.3 Contribution

The significant contributions of this work are listed as follows. 1) Data set in research can be shared as a case study; it can be provided as an actual data set for the subsequent research on the field of risk prediction in the financial and banking sector. 2) Research has defined a workflow template involved in a machine learning project from data collection to evaluation and finding the best suitable model on the collected data set. 3) Extensive experiments are carried out to compare the performances of the ensemble methods on the loan default prediction task based on a real dataset collected from VietCredit company. 4) Performance measures in previous studies that used Accuracy and ROC often did not reflect correctly in the imbalance data set. In this study, we used a more suitable measure, PR-AUC, for the unbalanced data set to avoid overly optimistic results when using other metrics.

Chapter 2 Data Collection and Preparation

2.1 Source Of The Dataset

The data set used to train loan default prediction models was collected from the following systems at VietCredit: 1.CAS - the system used to circulate and store customer information in the appraisal and approval of loan card registration documents. 2.PCB - the system to query and store customer credit information. 3.CMS - the system to manage card account information and transactions, fees, interest rates. 4.CMR - customer information storage system. 5.Scorecard – credit scoring system. The data collection diagram gives in Figure 1.

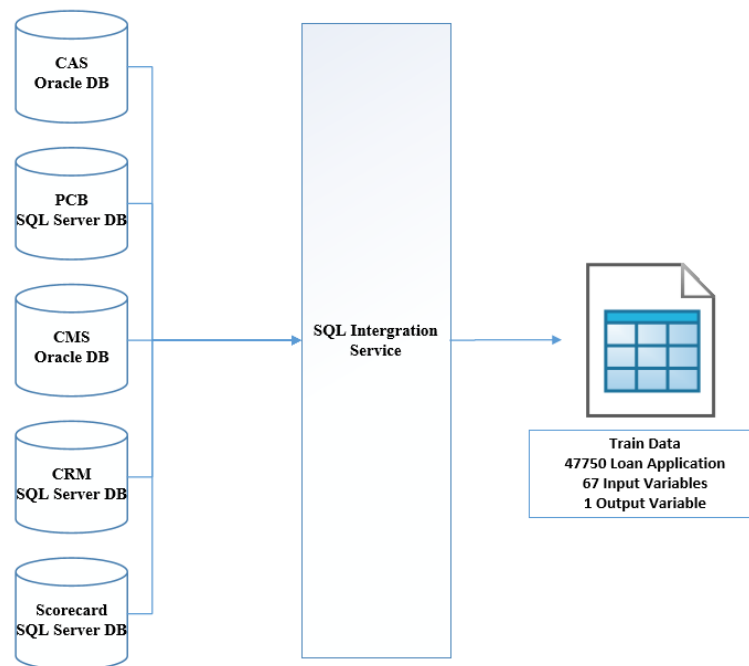


Figure 1: Data collection.

The data set has the 47750 loan applications (67 features) disbursement by the VietCredit platform from May 2020 to August 2020. Microsoft SQL Integration Service uses application-id attribute as key to join data from five transaction systems into a single CSV file. Figure 3 demonstrates the top ten rows and the first ten columns of the dataset.

Target	Appid	Sub_Industry	Employment_Status_Code	Salaried_Payment_Type	Age	Resident	Marital_Status	Address_Type	State	City	Dependents
1	91515	TNNH - LLP	JUNIOR OFFICER	LUONG CHUYEN KHOAN NGAN HANG KHONG BHXH - BANK TRANS	21	PARENTAL	Married	ResidentialAddress	No	HAU GIANG	0
1	91520	TNNH - LLP	JUNIOR OFFICER	LUONG CHUYEN KHOAN NGAN HANG KHONG BHXH - BANK TRANS	37	PARENTAL	Married	ResidentialAddress	No	VINH LONG	1
0	91523	TNNH - LLP	JUNIOR OFFICER	LUONG CHUYEN KHOAN NGAN HANG KHONG BHXH - BANK TRANS	31	SELF	Married	ResidentialAddress	No	VINH LONG	1
0	91607	NHA NUOC - STATE OWNED	GIAO VIEN	LUONG TIEN MAT CO BHXH - CASH SALARY WITH INSURANCE	29	SELF	Married	ResidentialAddress	No	DONG THAP	1
1	91668	TNNH - LLP	LAO DONG PHO THONG	LUONG CHUYEN KHOAN NGAN HANG CO BHXH - BANK TRANSFER	29	RENTED	Married	ResidentialAddress	No	BINH DUONG	0
1	91719	TNNH - LLP	JUNIOR OFFICER	LUONG CHUYEN KHOAN NGAN HANG CO BHXH - BANK TRANSFER	21	RENTED	Single	ResidentialAddress	No	THAI NGUYEN	0
1	91722	TNNH - LLP	LAO DONG PHO THONG	LUONG CHUYEN KHOAN NGAN HANG KHONG BHXH - BANK TRANS	25	PARENTAL	Single	ResidentialAddress	No	HA NOI	0
0	91730	NHA NUOC - STATE OWNED	JUNIOR OFFICER	LUONG CHUYEN KHOAN NGAN HANG CO BHXH - BANK TRANSFER	38	SELF	other	ResidentialAddress	No	HUNG YEN	0
1	91754	TNNH - LLP	JUNIOR OFFICER	LUONG CHUYEN KHOAN NGAN HANG CO BHXH - BANK TRANSFER	27	PARENTAL	Single	ResidentialAddress	No	DA NANG	0
0	91867	TNNH - LLP	TRUONG BO PHAN/TEAM LEADER	LUONG TIEN MAT CO BHXH - CASH SALARY WITH INSURANCE	24	PARENTAL	Single	ResidentialAddress	No	QUANG NAM	0

Figure 2: Training data sample.

During the study, we were consulted with some of the official experts of VietCredit and sorted out a wide array of essential variables. From the business perspective of our company, these variables are essential for credit scoring and loan default predictions. However, many of these variables may act differently in other companies in predicting loan defaults due to the difference in economic and business conditions. Table 3 below is a short explanation of each variable in the data set.

No	Variable	Description
1	Target	Outcome variable. Value 0 identity good customer who can pay off the loan. Otherwise, value 1
2	App_id	Loan application id
3	Sub_Industry	Type of company where customer work
4	Employment_Status_Code	Job title of customer
5	Salaried_Payment_Type	Type of salary payment
6	Age	Age of customer
7	Resident	Residence status
8	Marital_Status	Marital status
9	Address_Type	Address type
10	State	Yes – in Vietnam, No - Foreign
11	City	Place of issue of ID card
12	Dependents	Number of dependents
13	Children	Number of children
14	Nature_Business	Education level

No	Variable	Description
15	Company_Type	Type of business
16	Employment_type	Labor contract type
17	Employment_Status	Job title
18	Years_In_Current_Job	Number of year customer work at current company
19	Employer_Status	Company operation status
20	Employer_Establish_Day	Date of company establish
21	Years_In_Job	Total years of work experience
22	Maximum_Worststatus	Highest debt group in the last 12 months
23	Monthly_Payment	Monthly payment from other credit institutions (million VND)
24	Monthly_Income	Total monthly income (million VND)
25	Monthly_Income_Others	Total other source income/month (million VND)
26	Status3Month	Highest debt group in the last three months
27	Status6Month	Highest debt group in the last six months
28	Status12Month	Highest debt group in the last 12 months
29	Status24Month	Highest debt group in the last 24 months
30	NumberOfLiving	Number of contracts currently borrowed from other credit institutions
31	ResidualAmountSum	Total debt balance at other credit institutions
32	UnpaidDueInstalmentsAmountSum	Total unpaid amount at other credit institutions
33	NumberOfRequested	Number of inquiries at other credit institutions

No	Variable	Description
34	Ins_NumberOfLiving	The number of installments contracts currently borrowed from other credit institution
35	Ins_ResidualAmountSum	Total installments debt balance at other credit institutions
36	Ins_UnpaidDueInstalmentsAmountSum	The total unpaid amount of installments at other credit institutions
37	Ins_NumberOfRequested	The number of inquiries installments at other credit institution
38	CCard_NumberOfLiving	The number of card contracts currently borrowed from other credit institution
39	CCard_ResidualAmountSum	Total debt balance of card at other credit institutions
40	CCard_UnpaidDueInstalmentsAmountSum	The total unpaid amount of card at other Credit Institutions
41	CCard_NumberOfRequested	The number of inquiries of the card at other credit institution
42	Max_CL	The highest credit limit is granted at another credit institution
43	Card_sum_CL_Living	Total credit limit of card currently being granted at other credit institutions (Calculated on the valid contract)
44	Card_max_CL_Living	The highest credit limit of card issued at other credit institutions (Calculated on the valid contract)
45	Request_Limit	Request credit limit
46	Salary_Payment_Type_Code	Salary type
47	Occupation	
48	Industry	
49	Company_Type_Code	Code of company type (Vietcredit define)

No	Variable	Description
50	Gross_Monthly_Income	Total gross monthly income
51	Annual_Net_Income	Total annual net income
52	Annual_Gross_Income	Total annual gross income
53	Income_Other_Sources	Total income from other sources
54	Total_Monthly_Payment	Total monthly payment
55	Monthly_Expense	Total monthly expense
56	Income_Group	Group of income
57	Approve_Limit	Approve credit limit
58	First_Time_Borrowing_Date	First withdrawal date
59	Days_First_Time_Borrowing_Date	How many days from the first borrowing date to Aug 30, 2020
60	Total_Amount_Of_Borrowing_Money	The total amount of at first withdrawal date
61	Total_Amount_Of_Repayment_Money	The total amount paid by the customer
62	Last_Time_Repayment_Date	Latest repayment date
63	Days_Last_Time_Repayment_Date	How many days from the last repayment date to Aug 30, 2020
64	Total_Number_Of_Repayment	Number of repayments
65	Balance	Debt balance at August 31, 2020
66	Firstused_Balance	Debt balance at the time of the first withdrawal
67	Card_Type_Group	Request card type

Table 3: Variable description.

2.2 Analyze Data

The success of classification learning is heavily dependent on the quality of the data provided for training. This section will overview the loan default data set and perform an exploratory data

analysis to determine how to pre-process the data for improving the prediction result. Understanding the dataset can refer to several things, including but not limited to extracting important variables and leaving behind useless variables, identifying outliers, missing values, or human error, understanding the relationship(s), or lack of, between variables, ultimately maximizing the insights of a dataset and minimizing potential error later in the process.

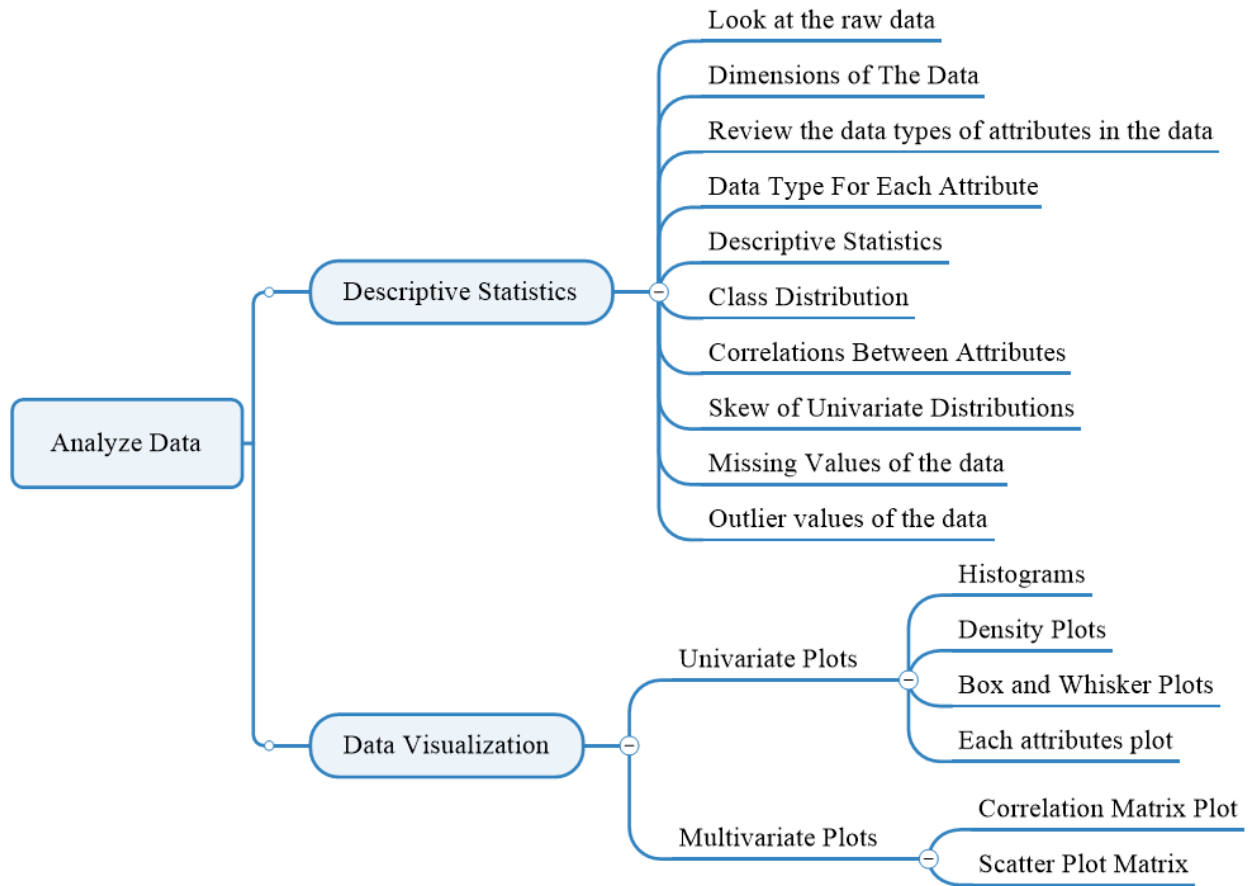


Figure 3: Exploratory data analysis.

Figure 3 is a guide to this study according to data analysis. The analysis begins with descriptive statistics in general to a detailed approach. Then the data will be visualized to have a multidimensional view of the data, giving us an initial orientation on the models to be applied.

2.3 Descriptive Statistics

2.3.1 Look At The Data

There is no substitute for looking at the raw data. Looking at the raw data can reveal insights things it cannot get any other way. It can also plant seeds that may later grow into ideas on how to better pre-process and handle the data for machine learning tasks. Figure 4 below shows the first ten rows of the data set.

Target	Appid	Sub_Industry	Employment_Status_Code	Salaried_Payment_Type	Age	Resident	Marital_Status	Address_Type	State	City	Dependents	Children	
0	1	91515	TNHH - LLP	JUNIOR OFFICER	LUONG CHUYEN KHOAN NGAN HANG KHONG BHXH - BANK...	21	PARENTAL	Married	ResidentialAddress	No	HAU GIANG	0	0
1	1	91520	TNHH - LLP	JUNIOR OFFICER	LUONG CHUYEN KHOAN NGAN HANG KHONG BHXH - BANK...	37	PARENTAL	Married	ResidentialAddress	No	VINH LONG	1	1
2	0	91523	TNHH - LLP	JUNIOR OFFICER	LUONG CHUYEN KHOAN NGAN HANG KHONG BHXH - BANK...	31	SELF	Married	ResidentialAddress	No	VINH LONG	1	1
3	0	91607	NHA NUOC - STATE OWNED	GIAO VIEN	LUONG TIEN MAT CO BHXH - CASH SALARY WITH INSU...	29	SELF	Married	ResidentialAddress	No	DONG THAP	1	0
4	1	91668	TNHH - LLP	LAO DONG PHO THONG	LUONG CHUYEN KHOAN NGAN HANG CO BHXH - BANK TR...	29	RENTED	Married	ResidentialAddress	No	BINH DUONG	0	0
5	1	91719	TNHH - LLP	JUNIOR OFFICER	LUONG CHUYEN KHOAN NGAN HANG CO BHXH - BANK TR...	21	RENTED	Single	ResidentialAddress	No	THAI NGUYEN	0	0
6	1	91722	TNHH - LLP	LAO DONG PHO THONG	LUONG CHUYEN KHOAN NGAN HANG KHONG BHXH - BANK...	25	PARENTAL	Single	ResidentialAddress	No	HA NOI	0	0
7	0	91730	NHA NUOC - STATE OWNED	JUNIOR OFFICER	LUONG CHUYEN KHOAN NGAN HANG CO BHXH - BANK TR...	38	SELF	other	ResidentialAddress	No	HUNG YEN	0	0
8	1	91754	TNHH - LLP	JUNIOR OFFICER	LUONG CHUYEN KHOAN NGAN HANG CO BHXH - BANK TR...	27	PARENTAL	Single	ResidentialAddress	No	DA NANG	0	0

Figure 4: Samples of data.

2.3.2 Dimensions of The Data

It is excellent to handle how much the data have, both in terms of rows and columns. Too many rows cause algorithms may take too long to train. Too few perhaps do not have enough data to train the algorithms. Too many features lead to some algorithms can be distracted or suffer poor performance due to the curse of dimensionality.

	Info	Value
0	Number of variables	67
1	Number of observations	47749
2	Missing cells	2221
3	Missing cells (%)	0.069424
4	Duplicate rows	0
5	Duplicate rows (%)	0
6	Total size in memory (Mbs)	24993.7
7	Average record size in memory (Bytes)	536.003

Figure 5: Shape of the data.

Figure 5 gives the dataset that has 47749 rows and 67 columns. Some missing cell values in the data set suggest using an imputer or remove approach on missing data. There is no number of duplicate rows; perhaps avoid overfitting when training model.

2.3.3 Data Type For Each Attribute

The type of each attribute is essential. Strings may need to be converted to floating-point values or integers to represent categorical or ordinal values. We can get an idea of the types of attributes by look at the raw data, as above. Figure 6 lists the data types of each attribute. Figure 7, and 8 shows the summary by data types in the data set.

Target	int64
Appid	int64
Sub_Industry	object
Employment_Status_Code	object
Salaried_Payment_Type	object
Age	int64
Resident	object
Marital_Status	object
Address_Type	object
State	object
City	object
Dependents	int64
Children	int64
Nature_Business	object
Company_Type	object
Employment_Type	object
Employment_Status	object
Years_In_Current_Job	float64
Employer_Status	object
Employer_Establish_Day	float64
Years_In_Job	float64
Maximum_Worststatus	int64
Monthly_Payment	float64
Monthly_Income	float64
Monthly_Income_Others	float64
Status3Month	int64

Figure 6: Datatypes of the data.

	Data Type	Count	Percentage
0	int64	31	46.3
1	object	23	34.3
2	float64	13	19.4

Figure 7: Summary data types.

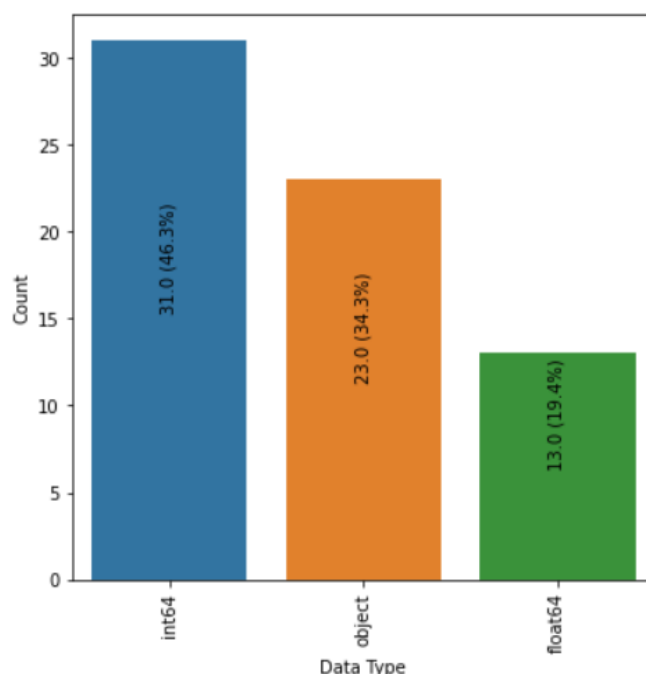


Figure 8: Data type histogram.

More than thirty percent of the attributes in the data set are the object data type, and the rest are numeric. Some algorithms can handle string properties independently, but most of them require conversion to a numeric type before training the model.

2.3.4 Descriptive Statistics

Descriptive statistics can give a great insight into the properties of each attribute. Often it can create more summaries than seeing by eye. The descriptive statistics list eight statistical properties of each attribute: Count, Mean, Standard Deviation, Minimum Value, 25th Percentile, 50th Percentile (Median), 75th Percentile Maximum Value. When describing the data this way, it is worth taking some time and reviewing observations from the results. Descriptive statistics might include the presence of NA values for missing data or surprising distributions for attributes.

	Target	Appid	Age	Dependents	Children	Years_In_Current_Job	Employer_Establish_Day	Years_In_Job	Maximum_Worststatus	Monthly_Payment	Monthly_Income
count	47749.000	47749.000	47749.000	47749.000	47749.000	47749.000	47749.000	47749.000	47749.000	47749.000	47749.000
mean	0.273	117806.978	29.701	0.424	-0.866	4.473	11.273	6.370	0.609	1.924	13.403
std	0.446	63937.029	7.553	0.676	101.693	4.694	7.194	5.649	0.510	30.363	652.260
min	0.000	1303.000	18.000	0.000	-11111.000	0.000	0.000	0.000	0.000	0.000	0.000
25%	0.000	62171.000	24.000	0.000	0.000	1.300	6.000	2.000	0.000	0.000	6.469
50%	0.000	123778.000	28.000	0.000	0.000	3.000	10.000	5.000	1.000	1.056	8.000
75%	1.000	172826.000	34.000	1.000	0.000	6.000	15.000	9.000	1.000	2.783	9.500
max	1.000	224513.000	55.000	8.000	11.000	38.000	412.000	45.000	6.000	6321.927	102560.670

Figure 9: Statistical summary of the data.

	Dependent Var	Var Name	Type	Distinct	Missing	Missing %	Outlier IQR	Outlier Zscore
0	x	Target	int64	2	0	0.000	0	0
1		Appid	int64	47749	0	0.000	0	0
2		Sub_Industry	object	10	2	0.004	0	0
3		Employment_Status_Code	object	16	0	0.000	0	0
4		Salaried_Payment_Type	object	10	0	0.000	0	0
5		Age	int64	38	0	0.000	954	305
6		Resident	object	7	0	0.000	0	0
7		Marital_Status	object	5	0	0.000	0	0
8		Address_Type	object	2	0	0.000	0	0
9		State	object	4	2	0.004	0	0
10		City	object	64	3	0.006	0	0
11		Dependents	int64	8	0	0.000	347	347
12		Children	int64	7	0	0.000	2298	4
13		Nature_Business	object	7	14	0.029	0	0
14		Company_Type	object	51	20	0.042	0	0
15		Employment_Type	object	7	2	0.004	0	0

Figure 10: Statistical each attribute.

Figure 9, and 10 shows the descriptive statistics of each feature in the data set. The statistical observations can reveal information about each attribute in the data. From there can be initial judgments about the following data processing sequence. For example, the min -11111 value in the Children attribute unreasonable and needs to be removed from the data set.

2.3.5 Class Distribution

On classification problems, one needs to know how balanced the class values are. Highly imbalanced problems (a lot more observations for one class than another) are common in actuality and may need special handling in many data preparation stages of the machine learning projects. In this study, we can see nearly triple the number of observations with class 0 (good in repayment) than with class 1 (wrong in repayment). Figure 11 shows the class distribution, and Figure 12 shows the imbalanced data set.

	Target	Count	Percentage
0	0	34697	72.7
1	1	13052	27.3

Figure 11: Class distribution of the data.

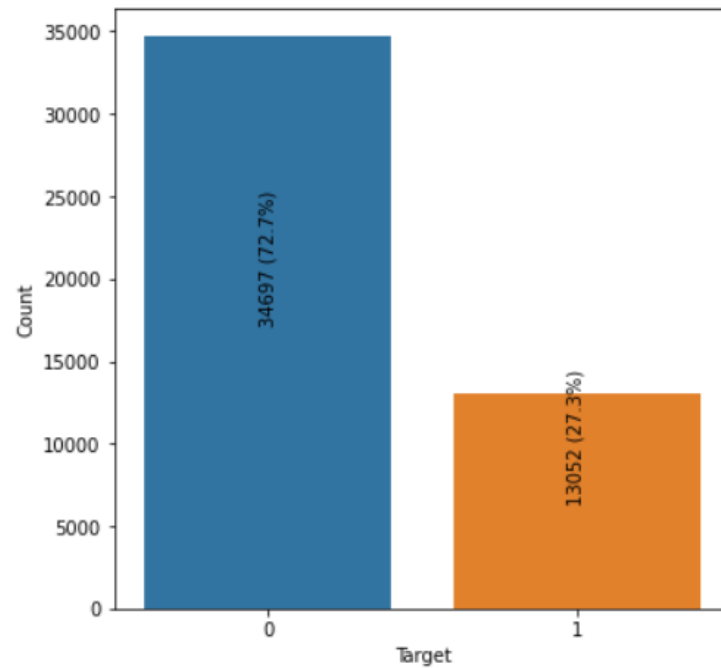


Figure 12: Imbalanced data set.

Figure 12 presents the data set of this research is imbalanced with 27.3 % of the minority class. The unbalanced training data requires some appropriate processing before the modeling, and it influences the choice of the suitable model or evaluation metrics.

2.3.6 Correlations Between Attributes

Correlation refers to the relationship between two variables and how they may or may not change together. The most common method for calculating correlation is Pearson's Correlation Coefficient, which assumes a normal distribution of the attributes involved. A correlation of -1 or 1 shows a full negative or positive correlation, respectively. Whereas a value of 0 shows no correlation at all. Some machine learning algorithms like linear and logistic regression can suffer poor performance if there are highly correlated attributes in the dataset. It is a good idea to review all of the pairwise correlations of the attributes in the dataset.

	Target	Appid	Age	Dependents	Children	Years_In_Current_Job	Employer_Establish_Day	Years_In_Job	Maximum_Worststatus	Monthly_Payment
Target	1.000e+00	0.042	-0.035	-0.021	4.493e-04	-0.067	-3.482e-02	-4.628e-02	0.008	9.465e-03
Appid	4.217e-02	1.000	0.011	-0.059	4.405e-03	0.034	-2.121e-02	-6.371e-02	0.054	4.048e-03
Age	-3.459e-02	0.011	1.000	0.267	-5.477e-03	0.558	1.169e-01	6.438e-01	0.127	1.681e-02
Dependents	-2.091e-02	-0.059	0.267	1.000	-3.915e-03	0.188	2.696e-02	2.422e-01	0.038	1.056e-02
Children	4.493e-04	0.004	-0.005	-0.004	1.000e+00	0.002	2.615e-03	4.965e-03	0.011	5.807e-04
Years_In_Current_Job	-6.680e-02	0.034	0.558	0.188	1.622e-03	1.000	2.028e-01	7.041e-01	0.096	1.697e-02
Employer_Establish_Day	-3.482e-02	-0.021	0.117	0.027	2.615e-03	0.203	1.000e+00	1.365e-01	0.030	5.196e-03
Years_In_Job	-4.628e-02	-0.064	0.644	0.242	4.965e-03	0.704	1.365e-01	1.000e+00	0.086	1.396e-02
Maximum_Worststatus	7.671e-03	0.054	0.127	0.038	1.095e-02	0.096	3.024e-02	8.600e-02	1.000	4.881e-02
Monthly_Payment	9.465e-03	0.004	0.017	0.011	5.807e-04	0.017	5.196e-03	1.396e-02	0.049	1.000e+00
Monthly_Income	-4.349e-03	0.009	0.007	-0.005	-3.299e-03	0.003	-2.489e-04	7.386e-04	-0.004	-2.522e-04

Figure 13: Correlations of attributes in the data.

Figure 13 lists all attributes across the top and down the side to correlate all pairs of attributes (twice because the matrix is symmetrical). We can see the diagonal line through the matrix from the top left to bottom right corners of the matrix shows the perfect correlation of each attribute with itself.

	Variable	Target
0	Firstused_Balance	0.207724
1	Ins_NumberOfRequested	0.197699
2	NumberOfRequested	0.174484
3	Approve_Limit	0.165456
4	Request_Limit	0.119952
5	Status24Month	0.057638
6	Ins_NumberOfLiving	0.051423
7	Years_In_Current_Job	-0.066805
8	Total_Amount_Of_Repayment_Money	-0.080628
9	Total_Amount_Of_Borrowing_Money	-0.142195
10	Balance	-0.256599
11	Total_Number_Of_Repayment	-0.553852

Figure 14: High correlations attributes with the output variable.

The data in Figure 14 filter out the correlation between the input and output variables with an absolute value ≥ 0.05 . The variables with the strongest negative correlation with the output variable are Total_Number_Of_Repayment with a value of -0.553852. It is clear that the larger the number of loan payments, the lower the likelihood of the loan default.

	Var 1	Var 2	Corr
0	Income_Other_Sources	Monthly_Income_Others	1.000000
1	Total_Monthly_Payment	Monthly_Payment	1.000000
2	CCard_ResidualAmountSum	ResidualAmountSum	1.000000
3	CCard_UnpaidDueInstalmentsAmount	UnpaidDueInstalmentsAmountSum	1.000000
4	Annual_Net_Income	Monthly_Income	1.000000
5	Monthly_Expense	Annual_Net_Income	0.999998
6	Monthly_Expense	Monthly_Income	0.999998
7	Annual_Net_Income	Gross_Monthly_Income	0.999995
8	Gross_Monthly_Income	Monthly_Income	0.999995
9	Monthly_Expense	Gross_Monthly_Income	0.999994
10	Total_Amount_Of_Repayment_Money	Total_Amount_Of_Borrowing_Money	0.970679
11	Status3Month	Maximum_Worststatus	0.953725

Figure 15: High correlation input variable pairwise.

Figure 15 shows a table of pairs of input variables with near absolute correlation. If not removed, a pair of highly correlated input variables can affect model performance. For example, the input variable pair Income_Other_Source and Monthly_Income_Others have an absolute correlation. By observing the original database, we find that the two variables are duplicates. In this study, variables in the Var2 column will be pre-removed during data cleaning.

2.3.7 The skew of Univariate Distributions

Skew refers to a distribution that is assumed Gaussian (standard or bell curve) that is shifted or squashed in one direction or another. Many machine learning algorithms assume a Gaussian distribution. Knowing that an attribute has a skew may need to perform data preparation to correct

the skew and later improve the accuracy of models. The skew results show a positive (right) or negative (left) skew. Values closer to zero show no skew.

Target	1.017
Appid	-0.179
Age	0.889
Dependents	1.538
Children	-109.246
Years_In_Current_Job	2.282
Employer_Establish_Day	5.884
Years_In_Job	1.716
Maximum_Worststatus	0.420
Monthly_Payment	193.321
Monthly_Income	133.136
Monthly_Income_Others	40.131
Status3Month	-0.257
Status6Month	1.501
Status12Month	3.037
Status24Month	3.064
NumberOfLiving	1.336
ResidualAmountSum	11.750
UnpaidDueInstalmentsAmountSum	51.436
NumberOfRequested	3.195
Ins_NumberOfLiving	1.250
Ins_ResidualAmountSum	0.000
Ins_UnpaidDueInstalmentsAmountSu	0.000
Ins_NumberOfRequested	2.648
CCard_NumberOfLiving	4.198
CCard_ResidualAmountSum	11.750
CCard_UnpaidDueInstalmentsAmount	51.436
CCard_NumberOfRequested	11.896
Max_CL	13.776
Card_Sum_CL_Living	12.290
Card_Max_CL_Living	7.917
Request_Limit	2.637
Gross_Monthly_Income	133.135
Annual_Net_Income	133.136
Annual_Gross_Income	218.052
Income_Other_Sources	40.131
Total_Monthly_Payment	193.321
Monthly_Expense	133.139
Approve_Limit	4.830
Total_Amount_Of_Borrowing_Money	53.229
Total_Amount_Of_Repayment_Money	55.995
Total_Number_Of_Repayment	0.757
Balance	0.006
Firstused_Balance	3.080
dtype: float64	

Figure 16: Skew of attribute distributions in the data

Some of the attributes in the data are highly skewed. For example, Children is 109, Monthly_Payment is 193, Annual_Gross_Income is 219. The skew of data fields can indicate anomalies in the data set; it needs actions such as removing the outlier or using transform algorithms.

2.3.8 Missing Values Of The Data

The study's data set has missing values, which can cause problems for many machine learning algorithms. Data can have missing values for several reasons, such as observations not recorded or error typing. Handling missing data is essential as many machine learning algorithms do not support data with missing values. It is good practice to identify and replace missing values for each column in the input data before modeling the prediction task. It is called missing data imputation or imputing for short. A popular approach for data imputation calculates a statistical value for each column (such as a mean) and replacing all missing values for that column with the statistic. It is a popular approach because the statistic is easy to calculate using the training dataset and because it often results in a good performance.

	Var Name	Missing	Missing %
0	Balance	1943	4.069
1	First_Time_Borrowing_Date	80	0.168
2	Firstused_Balance	80	0.168
3	Last_Time_Repayment_Date	44	0.092
4	Income_Group	28	0.059
5	Company_Type	20	0.042
6	Nature_Business	14	0.029
7	City	3	0.006
8	Sub_Industry	2	0.004
9	State	2	0.004
10	Employment_Type	2	0.004
11	Industry	2	0.004
12	Employment_Status	1	0.002

Figure 17: Missing values of attributes.

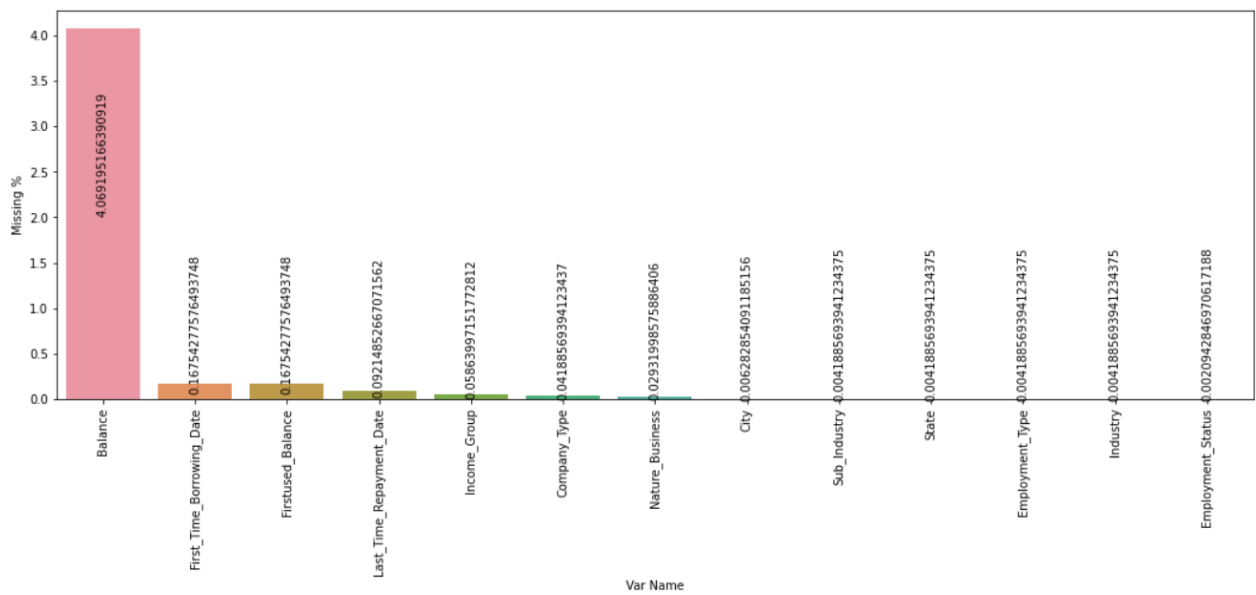


Figure 18: Percentage missing values of attributes.

Figures 17 and 18 give a statistical report about missing values in each attribute. The rate of missing data in the training data set is relatively low, so in this study, the missing data will be eliminated without worrying about loss of information.

2.3.9 Outlier Values Of The Data

When modeling, it is essential to clean the data sample to ensure that the observations best represent the problem. Some attributes in the dataset can contain extreme values outside the range of what is expected. These are called outliers, and often machine learning modeling and model skill, in general, can be improved by understanding and even removing these outlier values. Many machine learning algorithms are sensitive to the range and distribution of attribute values in the input data. Outliers in input data can skew and mislead the training process of machine learning algorithms resulting in longer training times, less accurate models, and ultimately more poor results. Even before predictive models are prepared on training data, outliers can result in misleading representations and, in turn, misleading interpretations of collected data. Outliers can skew the summary distribution of attribute values in descriptive statistics like mean and standard deviation and plots such as histograms and scatterplots, compressing the body of the data.

	Var Name	Outlier IQR	Outlier Zscore
0	CCard_NumberOfRequested	10006	901
1	Approve_Limit	9260	1412
2	Ins_NumberOfRequested	6177	1170
3	Card_Sum_CL_Living	5365	907
4	Card_Max_CL_Living	5365	1183
5	CCard_NumberOfLiving	5363	975
6	Request_Limit	5345	1045
7	ResidualAmountSum	4637	952
8	CCard_ResidualAmountSum	4637	952
9	Gross_Monthly_Income	4222	3
10	Total_Amount_Of_Repayment_Money	4189	400
11	Monthly_Income_Others	3542	534
12	Total_Amount_Of_Borrowing_Money	3439	390
13	Annual_Gross_Income	3420	1
14	Income_Other_Sources	3228	534
15	Years_In_Current_Job	2589	1089
16	Children	2298	4
17	Monthly_Income	2264	3
18	Annual_Net_Income	2264	3
19	Monthly_Expense	2173	3
20	Years_In_Job	1991	914
21	NumberOfRequested	1773	1108
22	Status24Month	1427	912

Figure 19: Outlier of each attribute.

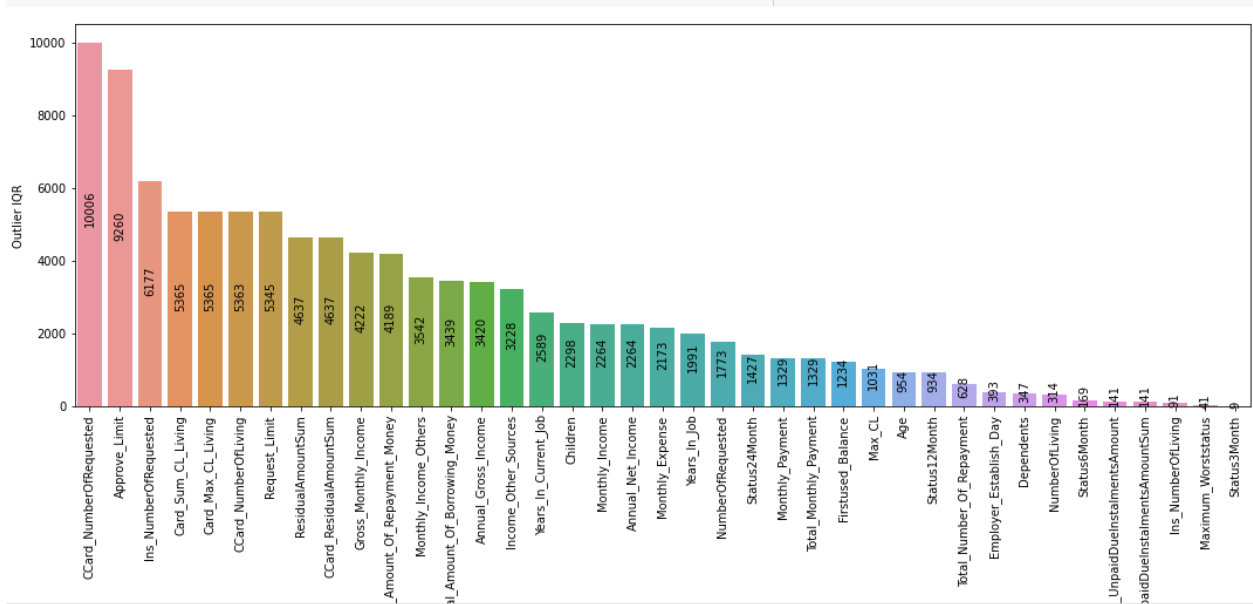


Figure 20: Outlier histogram of each attribute.

Figure 19, and 20 shows outlier values of each attribute in the data set. This study intends to treat outliers by reducing the scale of numerical features to the interval $[0, 1]$ range. Hence, all numerical properties are of the same scale, which increases the training efficiency and the prediction of the model.

2.4 Data Visualization

Sometimes, data does not make sense until looking at it in a visual form, such as charts and plots. To understand the data to get the best results from machine learning algorithms, the fastest way to learn more about the data is to use data visualization. Visualizing a data set is an important skill both in applied statistics and applied machine learning. In this study, the five types of plots were used to help better understand the collected data.

2.4.1 Analyze Univariate With Histogram Plots

A fast way to get an idea of the distribution of each attribute is to look at histograms. Histograms group data into bins and provide a count of the number of observations in each bin. From the shape of the bins, we can quickly get a feeling for whether an attribute is Gaussian, skewed, or even has an exponential distribution. It can also help to see possible outliers.

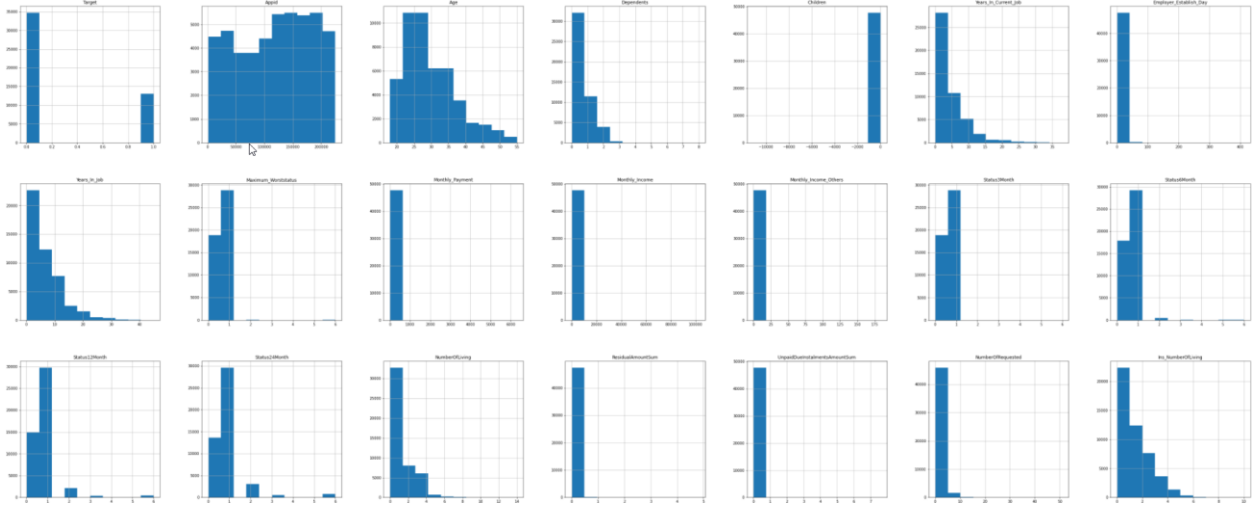


Figure 21: Histograms of each attribute.

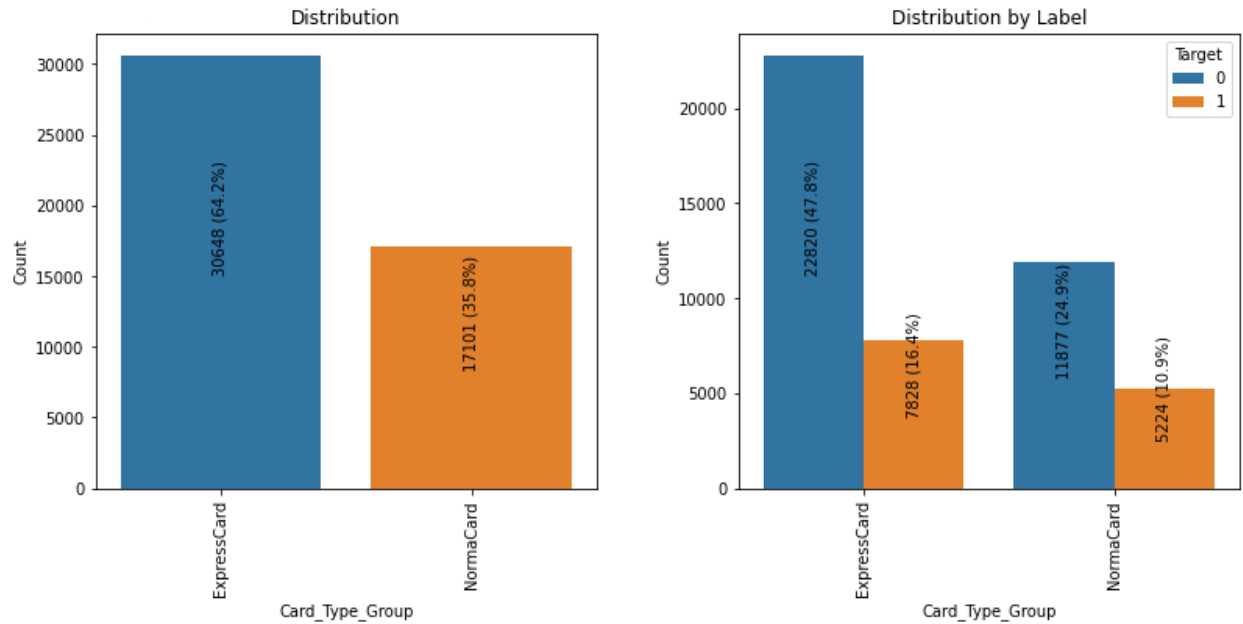


Figure 22: Histogram of category attribute.

Figure 21 plots a histogram of each numeric attribute in the data set. Look at the histogram plot of the data set of this study, and we can see that perhaps the attributes `Card_Expiry`, `Ins_NumberOfLiving` may have an exponential distribution. We can also see that perhaps the `Age` and `Balance` attributes may have a Gaussian or nearly Gaussian distribution. The histogram is interesting because many machine learning techniques assume a Gaussian univariate distribution on the input variables. Figure 22 shows an example of the histogram of category properties. The

figure on the left shows the distribution of values in the attribute. The figure on the right shows the distribution of data by the dependent variable.

2.4.2 Analyze Univariate With Density Plots

Density plots are another way of getting a quick idea of the distribution of each attribute. The plots look like an abstracted histogram with a smooth curve drawn through the top of each bin. It much like do with the histograms, but the distribution for each attribute is more straightforward than the histograms. Figure 23 is an example of a density plot of some attribute in the data set.

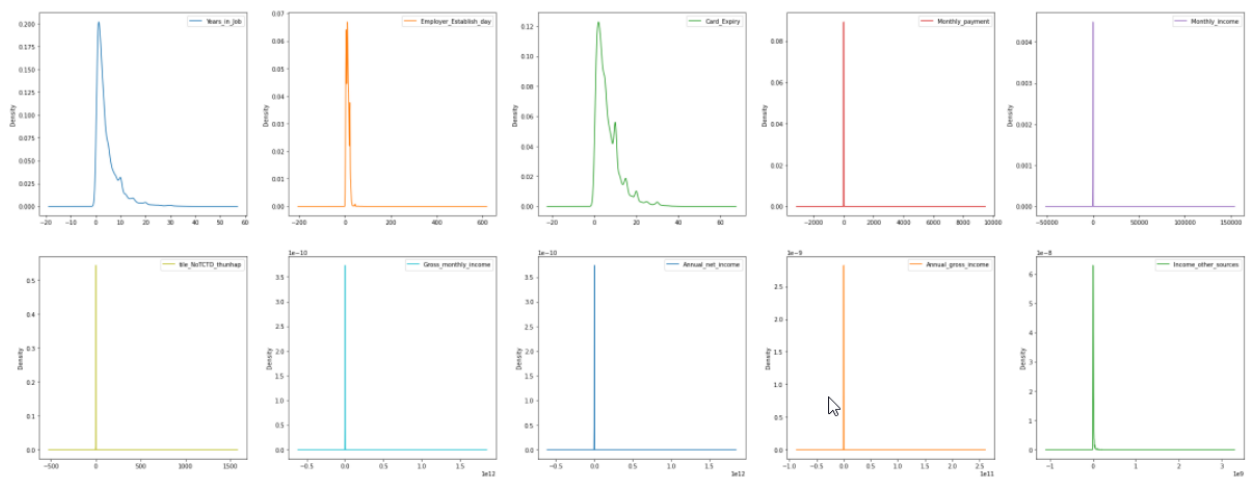


Figure 23: Density plot of each attribute.

2.4.3 Analyze Univariate With Box Plots

Another helpful way to review the distribution of each attribute is to use Box and Whisker Plots or boxplots for short. Boxplots summarize the distribution of each attribute, drawing a line for the median (middle value) and a box around the 25th and 75th percentiles. The whiskers give an idea of the spread of the data, and dots outside of the whiskers show candidate outlier values (values that are 1.5 times greater than the size of spread of the middle 50% of the data).

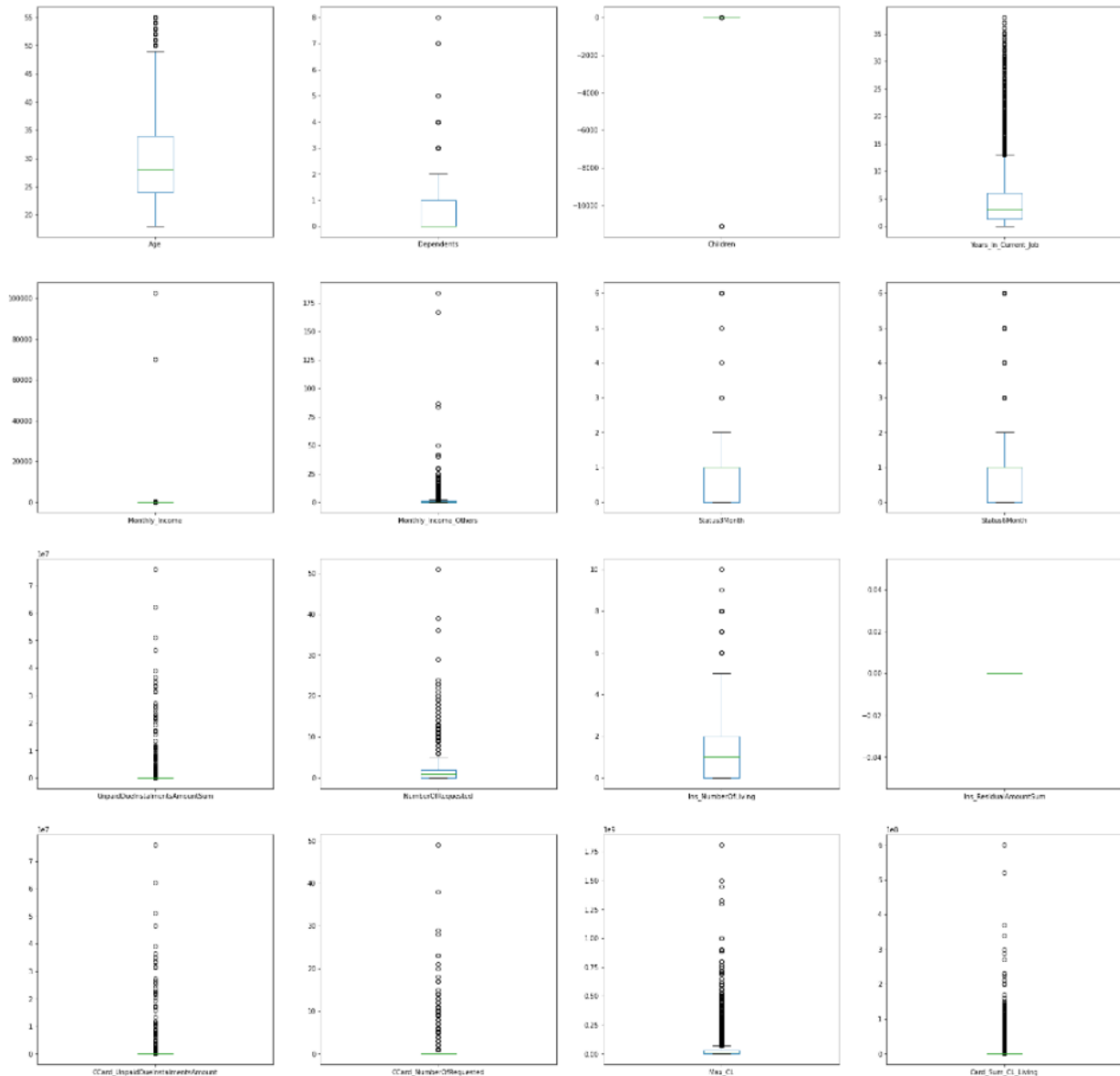


Figure 24: Box plot of each attribute.

Figure 24 below shows that the spread of attributes is quite different. Some like Dependents, Year_In_Job appear quite skewed towards smaller values.

2.4.4 Analyze Multivariate With Correlation Matrix Plot

Correlation indicates how related the changes are between two variables. If two variables change in the same direction, they are positively correlated. If they change in opposite directions together (one goes up, one goes down), they are negatively correlated. We can calculate the correlation between each pair of attributes. It is called a correlation matrix. We can then plot the correlation

matrix and get an idea of which variables have a high correlation. Correlation Matrix Plot is helpful because some machine learning algorithms like linear and logistic regression can have poor performance if there are highly correlated input variables in the data. We can see that the matrix is symmetrical, i.e., the bottom left of the matrix is the same as the top right. It is helpful as we can see two different views on the same data in one plot. We can also see that each variable is perfectly positively correlated with itself (as you would have expected) in the diagonal line from top left to bottom right.

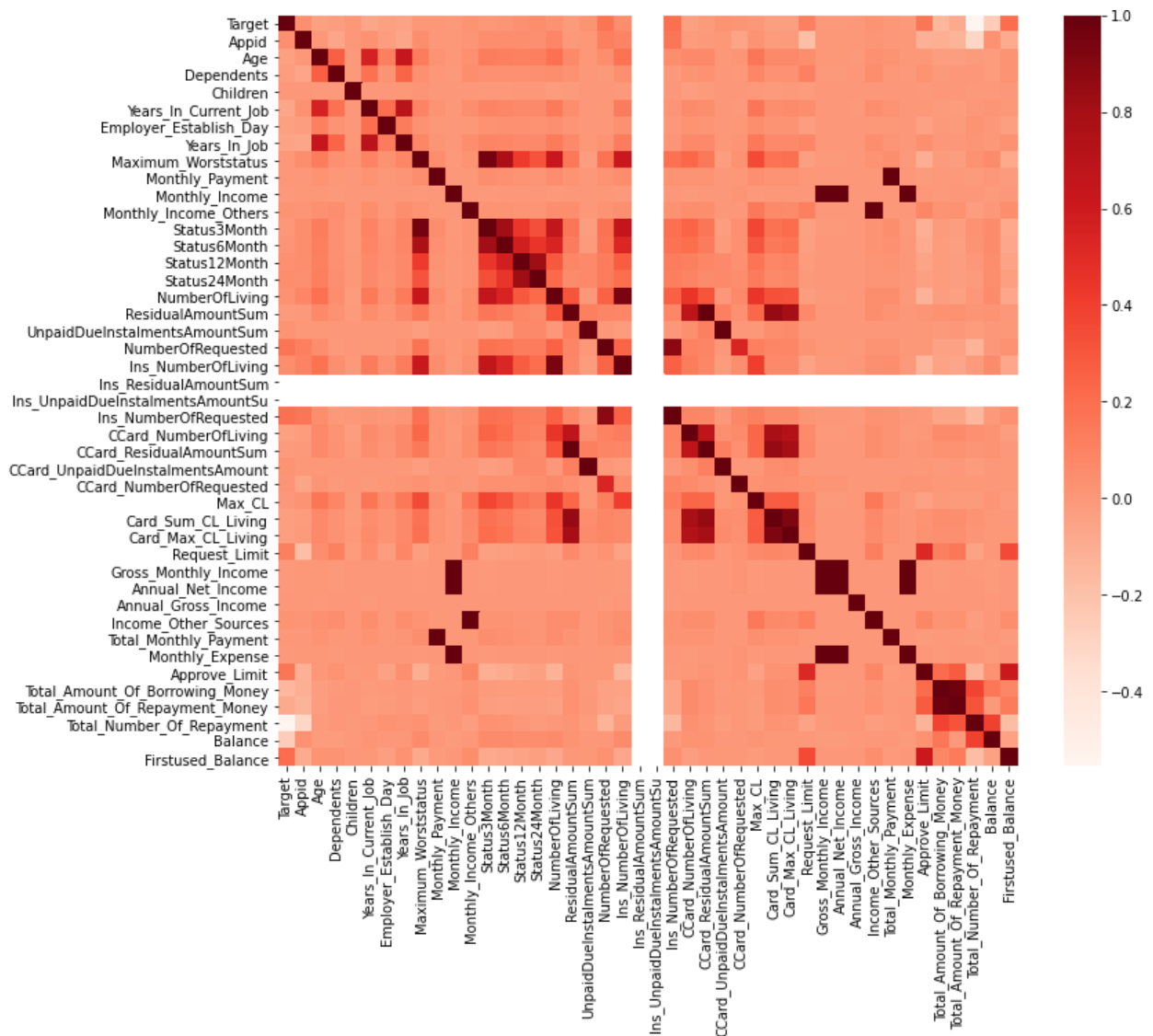


Figure 25: Correlation matrix plot.

Figure 25 shows the correlation matrix among the variables in the data. The main diagonal represents the relation correlated with the variable itself. Therefore, the analysis only needs to be

based on the upper triangle or the lower triangle. Darker colors indicate a higher correlation. For example, the correlation value between Monthly_Income and Gross_Monthly_Income is 1 representing absolute correlation.

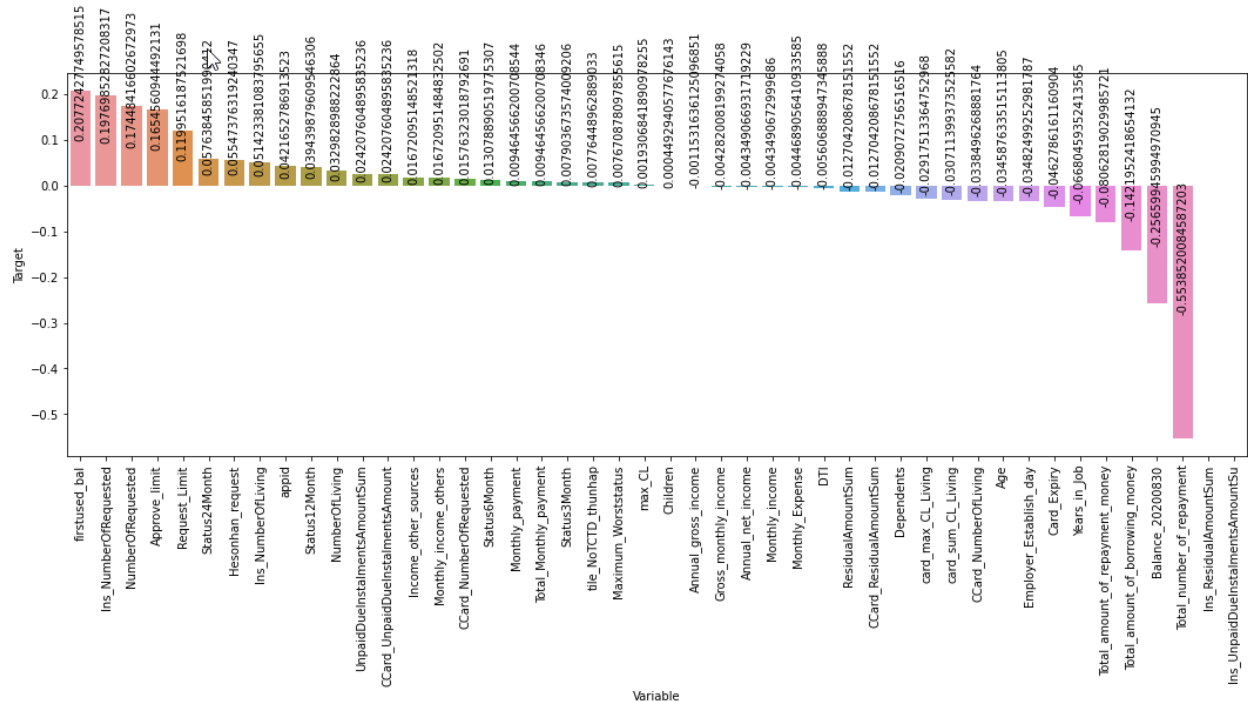


Figure 26: Correlation matrix with output plot.

2.4.5 Analyze Multivariate With Scatter Matrix Plot

A scatter plot shows the relationship between two variables as dots in two dimensions, one axis for each attribute. We can create a scatter plot for each pair of attributes in the data. Drawing all these scatter plots together is called a scatter plot matrix. Scatter plots help spot structured relationships between variables, like whether we could summarize the relationship between two variables with a line. Attributes with structured relationships may also be correlated and suitable candidates for removal from the dataset. Like the Correlation Matrix Plot above, the scatter plot matrix is symmetrical. A Scatter plot is helpful to look at the pairwise relationships from different perspectives. Because there is little point in drawing a scatter plot of each variable, the diagonal shows histograms of each attribute.

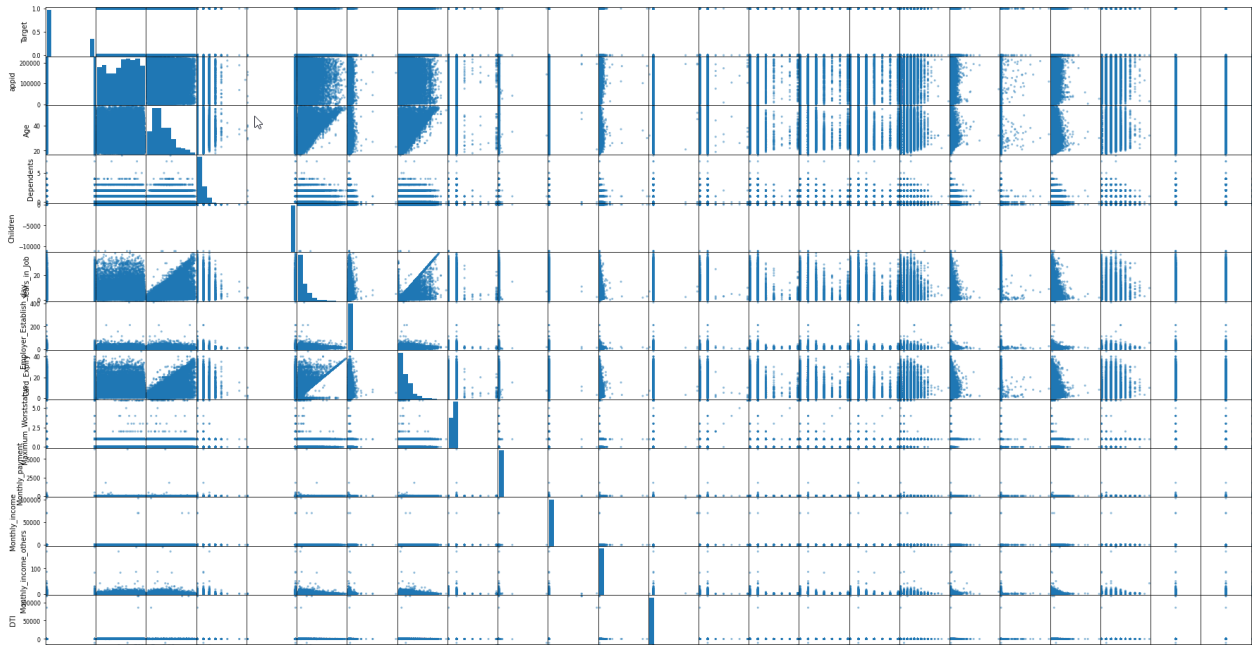


Figure 27: Scatter plot matrix of the data.

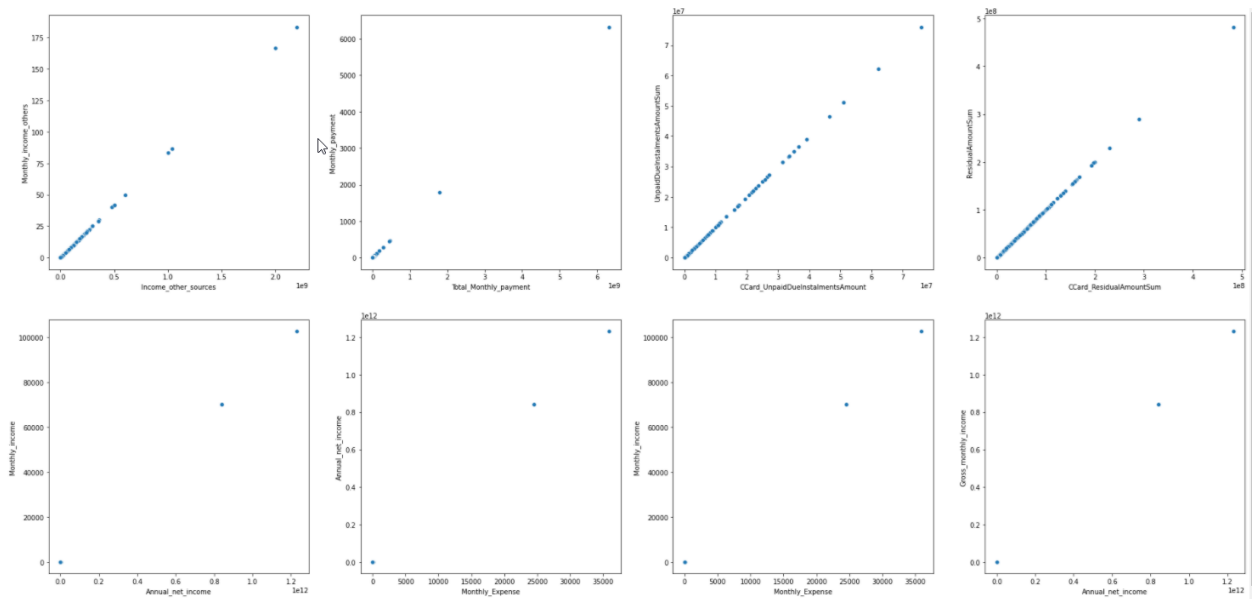


Figure 28: Scatter plot matrix of high correlation attributes.

As Figure 28 shows, in the training data, there are many duplicate pairs of data variables. The duplication can be caused by input data in the source data but not detected during data collection. Some pairs of variables are absolute linear relations, so one of two will be removed from the data set.

2.5 Prepare Data

Many machine learning algorithms make assumptions about the data. It is often a perfect idea to prepare the data in such a way to best expose the structure of the problem to the machine learning algorithms that are intended to use. The prepared data is the outcome of exploratory data analysis. In this study, data cleaning and data transform were performed, such as the flow shows in Figure 29.

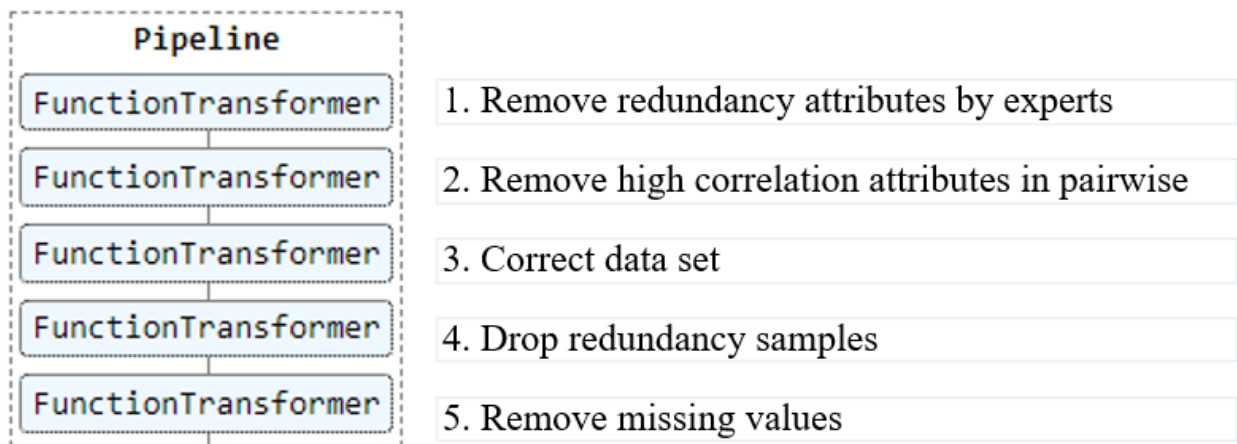


Figure 29: Prepare data pipeline.

Steps 1 to 5 perform cleaning the data, such as removing redundancy attributes and samples, correcting data, and removing missing values. The data after cleaning will remain 45670 rows and 45 columns.

Attributes to delete	Reason to delete
App_id	Unique application index
Employment_Status_Code	Duplicate with "Employment_status"
Address_Type	Not to use as explanatory variables since nearly 100% of the value is the same one
Children	Not to use as explanatory variables since nearly 100% of the value is the same one
Employer_Status	Not to use as explanatory variables since nearly 100% of the value is the same one
Ins_ResidualAmountSum	Only one value

Ins_UnpaidDueInstalmentsAmountSu	Only one value
Occupation	Not to use as explanatory variables since nearly 100% of the value is the same one
Industry	Not to use as explanatory variables since nearly 100% of the value is the same one
Company_Type_Code	Duplicate with "Company_type"
First_Time_Borrowing_Date	Not available at the time of inspection. Use Days_First_Time_Borrowing_Date instead.
Last_Time_Repayment_Date	Not available at the time of inspection. Use Days_ Last_Time_Repayment_Date instead.

Table 4: Redundancy attributes by experts.

Attributes to delete	Correlation attributes	Correlation
Monthly_Income_Others	Income_Other_Sources	1
Monthly_Payment	Total_Monthly_Payment	1
ResidualAmountSum	CCard_ResidualAmountSum	1
UnpaidDueInstalmentsAmountSum	CCard_UnpaidDueInstalmentsAmount	1
Monthly_Income	Annual_Net_Income	1
Annual_Net_Income	Monthly_Expense	0.999998
Monthly_Income	Monthly_Expense	0.999998
Gross_Monthly_Income	Annual_Net_Income	0.999995
Monthly_Income	Gross_Monthly_Income	0.999995
Gross_Monthly_Income	Monthly_Expense	0.999994
Total_Amount_Of_Borrowing_Money	Total_Amount_Of_Repayment_Money	0.970679
Maximum_Worststatus	Status3Month	0.953725

Table 5: Redundancy attributes correlation pairwise.

Table 4 lists redundancy attributes observed by experts in VietCredit. The reasons to delete perhaps duplicate with another or attributes have the same one value. Table 5 identifies pairwise attributes with nearly absolute or absolute correlation values so that one of them will remove from the data set before modeling. As shown in the Descriptive Statistics section above, the data set has both string (category) and numeric (int, float) data type. The way used to convert the category to number data type was one hot encoder. The numeric attributes have different scales going to be rescaled before modeling. After data pre-processing, the dimension of the data set down from (47749, 67) to (45670, 45) in rows and columns.

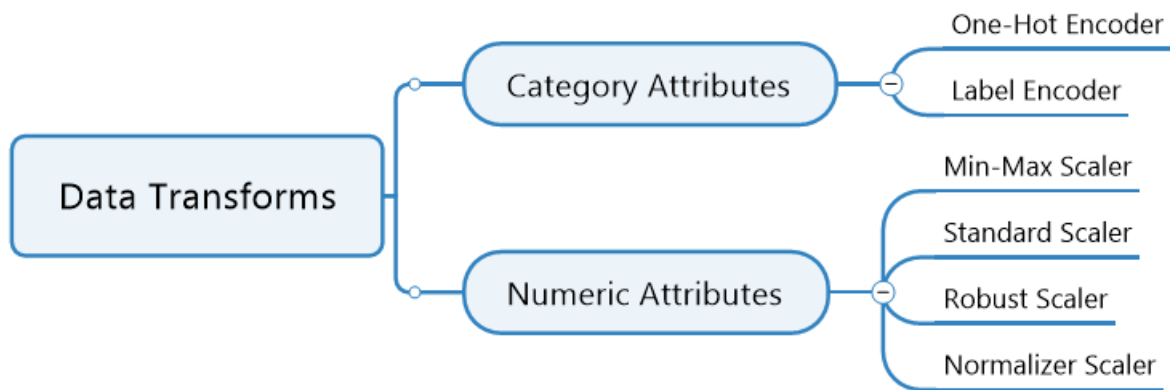


Figure 30: Data transforms approach.

Chapter 3 Implementation and Analysis

3.1 Methodology

The overview to detail and data-first approach used to process works in the thesis, and the practice approach will be focused more than underlying theory algorithms study. A Structured Step-By-Step Process is used to organize entire work procedures into key divisions, shown in Figure 1. The objective research was defined at the first stage of the study to define problems in predicting credit repayment ability. Some background knowledge and the business domain were investigated and advised by domain experts of the finance field. After that, the data set collected from the transaction systems, analyzed and visualized to figure out all insights' perspectives aim to understand the data better. A data pre-processing was applied to transform the data met some requirements of algorithms. After that, the processed data evaluated with multiple numbers kind of Supervised machine learning models. The algorithms turning and ensemble method also was used to improve the performance of models. The Precision and Recall (AUC) metric was selected to evaluate the performance of models to avoid imbalanced challenges in the data set. Finalize the best model was selected and used to predict the held-out evaluation data.

3.2.1 Machine Learning Workflow

The machine learning process is not linear but cyclic. The loop between steps could occur much time and probably spend most of the time in tight loops between steps 3-4 or 3-4-5 until achieving a sufficient level of accuracy. Figure 31 shows a machine learning workflow in this study.

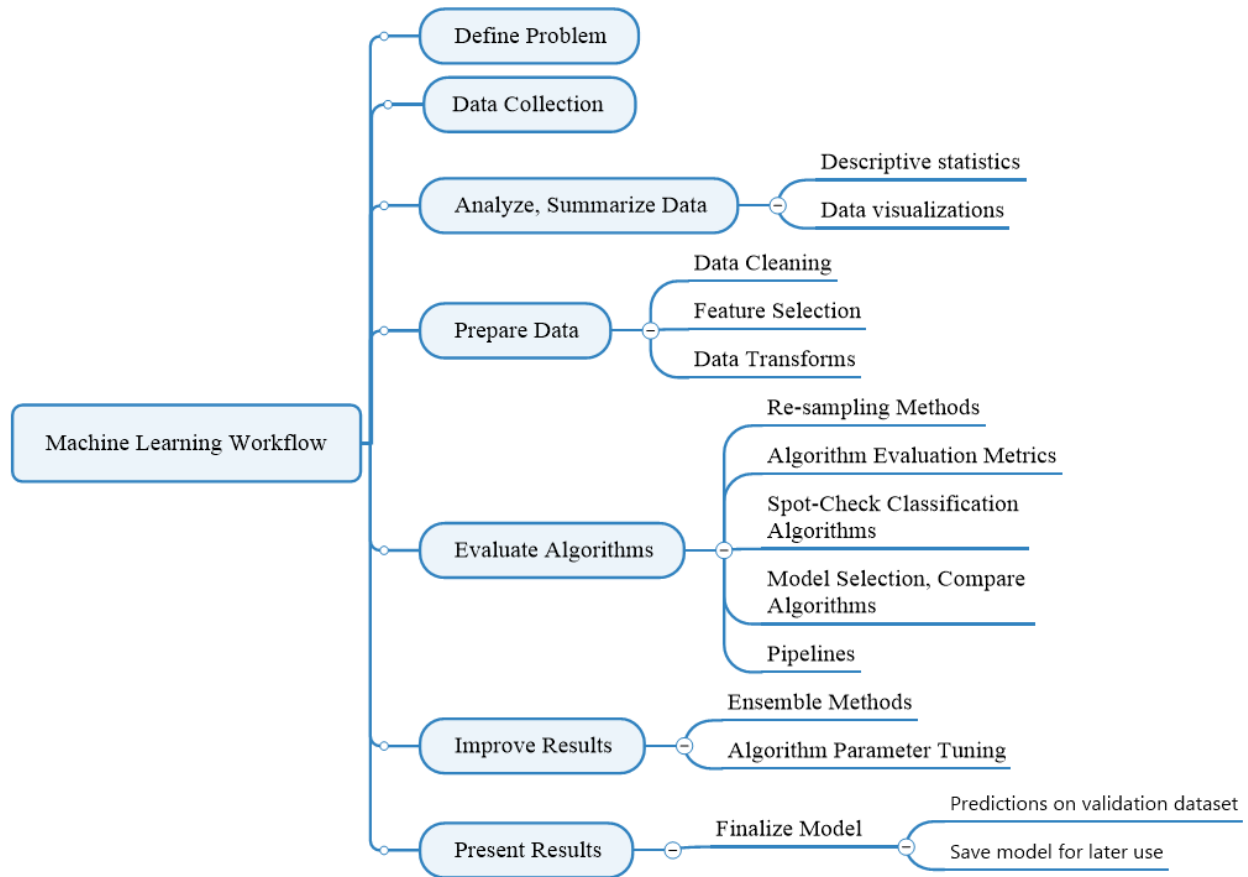


Figure 31: Machine learning workflow.

Define Problem: Investigate and characterize the problem to better understand the goals of the project. The focus of this project will be the VietCredit Normal Card Probability Default dataset. The problem is to predict the bad or good loan application of customers from that data. Each sample had a set of 67 attribute values include numeric, string type. The label associated with each record contains the number 1 if the loan application is wrong or probability default; otherwise is 0. Class distribution is 73% for bad and 27% for good, and this is a skew distribution cause imbalanced data challenge to classification models.

Data Collection: The data set was collected from four transaction systems of VietCredit.

Analyze Data: Use descriptive statistics and visualization to understand better the data which we have available.

Prepare Data: Use data transforms, impute missing values, remove outliers, to better expose the structure of the prediction problem to modeling algorithms.

Evaluate Algorithms: Select the right score metric based on data set structure, design a test to evaluate several standard algorithms on the data, and select the top few to investigate further.

Improve Results: Use hyper-parameters algorithm tuning and ensemble methods to get the most out of well-performing algorithms on the data set.

Present Results: Finalize the model, make predictions, and present results.

3.2.2 Evaluate Algorithms

This step is about finding a subset of machine learning algorithms that are good at exploiting the data set structure (e.g., have better than average skill). Evaluate Algorithms involves steps such as: Separating a validation dataset to use for later confirmation of the skill of the developed model and defining test options such as cross-validation and the evaluation metric to use. Spot-checking a suite of linear and nonlinear machine learning algorithms. It is comparing the estimated accuracy of algorithms. Figure 32 is a diagram that includes some algorithms we intend to try in this research.

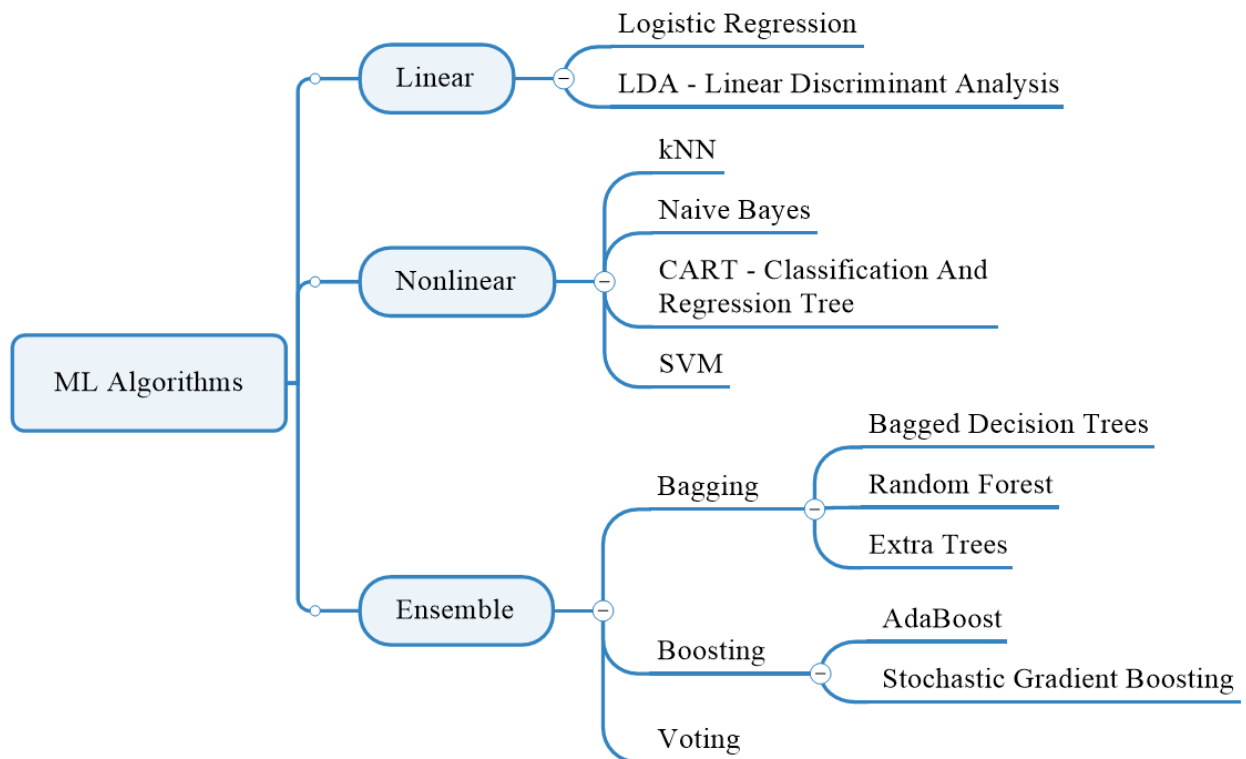


Figure 32: Machine Learning Algorithms

3.2 Validation Dataset

To avoid overfitting, it is a good idea to use a validation hold-out set. The hold data set was used at the end of the project to confirm the accuracy of the final model. It is test data that can help give confidence in the estimates of accuracy on unseen data. This study will use 80% of the dataset for modeling and hold back 20% for validation.

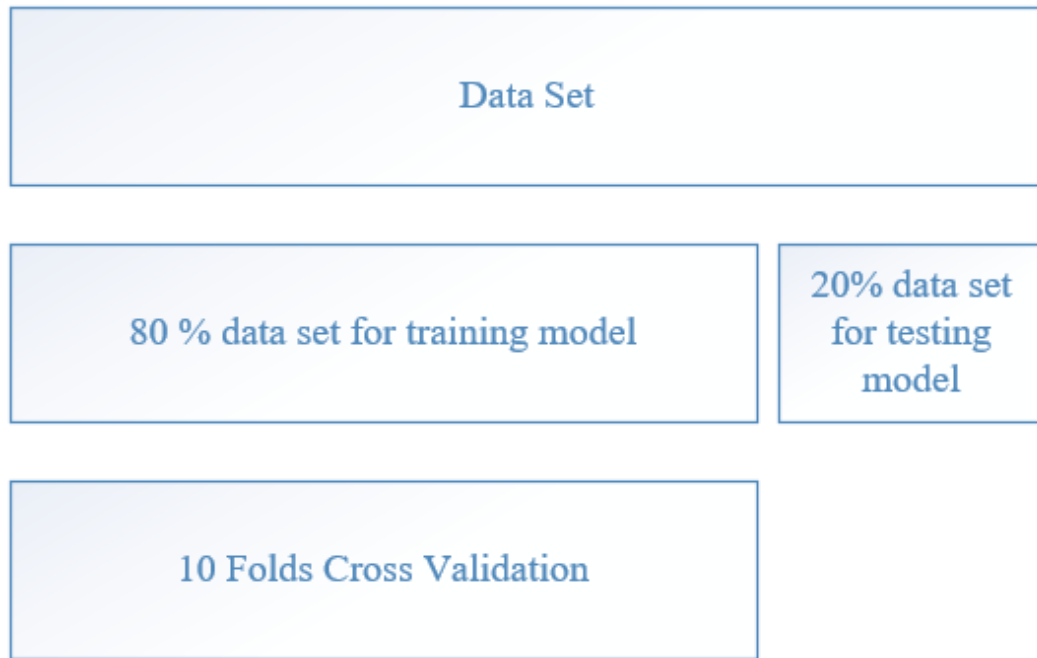


Figure 33: Validation dataset.

Data after cleaning will be divided into two parts with a ratio of 80:20. 20% will be used to check the accuracy of the best-chosen model. The remaining 80% will be divided into ten folds used to train the model using cross-validation.

3.3 Evaluation Metrics

A classifier is only as good as the metric used to evaluate it. If the wrong metric is chosen to evaluate models, it is likely to choose an imperfect model, or in the worst case, be misled about the expected performance of models. Choosing an appropriate metric is generally challenging in applied machine learning but is particularly difficult for imbalanced classification problems because most of the standard metrics that widely used assume a balanced class distribution. The data set of this research is imbalanced, with 27% target variable values are 1 (bad in repayment),

and otherwise is 73%. For imbalanced classification problems, the majority class is typically referred to as the negative outcome, and the minority class is typically referred to as the positive outcome. Both the precision and the recall are focused on the positive class (the minority class) and are unconcerned with the true negatives (majority class). Precision-recall curves and PR AUC are recommended for highly skewed domains, so it is used in this study to evaluate the model's performance. This metric avoids the use of true negative in the calculation, so it avoids an excessively optimistic view of ROC curves' performance.

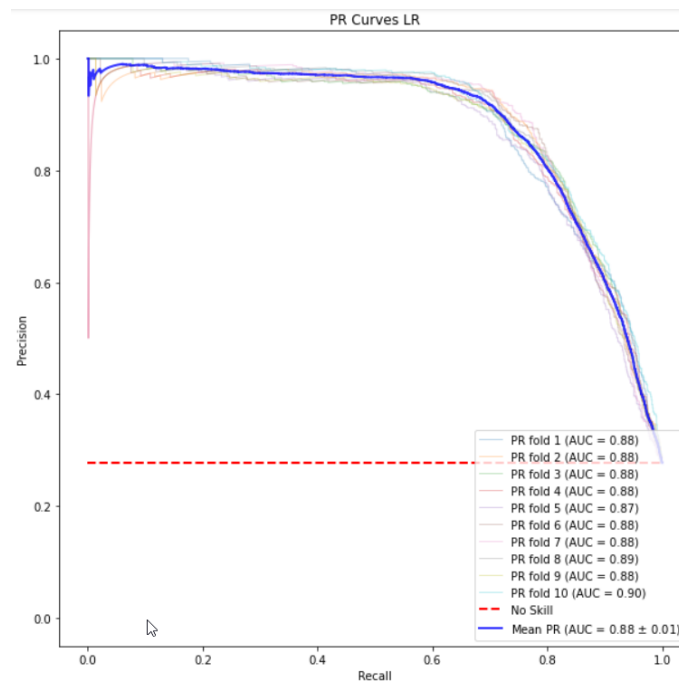


Figure 34: Precision-Recall AUC example.

A model with perfect skill is depicted as a point at a coordinate of (1,1). A skillful model is represented by a curve that bows towards a coordinate of (1,1). A no-skill classifier will be a horizontal line on the plot with a precision that is proportional to the number of positive examples in the dataset. For a balanced dataset, this will be 0.5. The focus of the PR curve on the minority class makes it an effective diagnostic for imbalanced binary classification models. Precision-recall curves (PR curves) are recommended for highly skewed domains where ROC curves may provide an excessively optimistic view of the performance.

3.4 Evaluate Algorithms: Baseline

This study selects a suite of different algorithms capable of working on this classification problem. The six algorithms selected include Linear Algorithms: Logistic Regression (LR) and Linear Discriminant Analysis (LDA). Nonlinear Algorithms: Classification and Regression Trees (CART), Support Vector Machines (SVM), Gaussian Naive Bayes (NB), and k-Nearest Neighbors (KNN). The algorithms all use default tuning parameters. The category attribute will transform by using the one-hot encoder. The numeric attributes were original and not transformed. The mean and standard deviation of Precision-Recall AUC on all folds for each algorithm was calculated and used to compare models' performance.

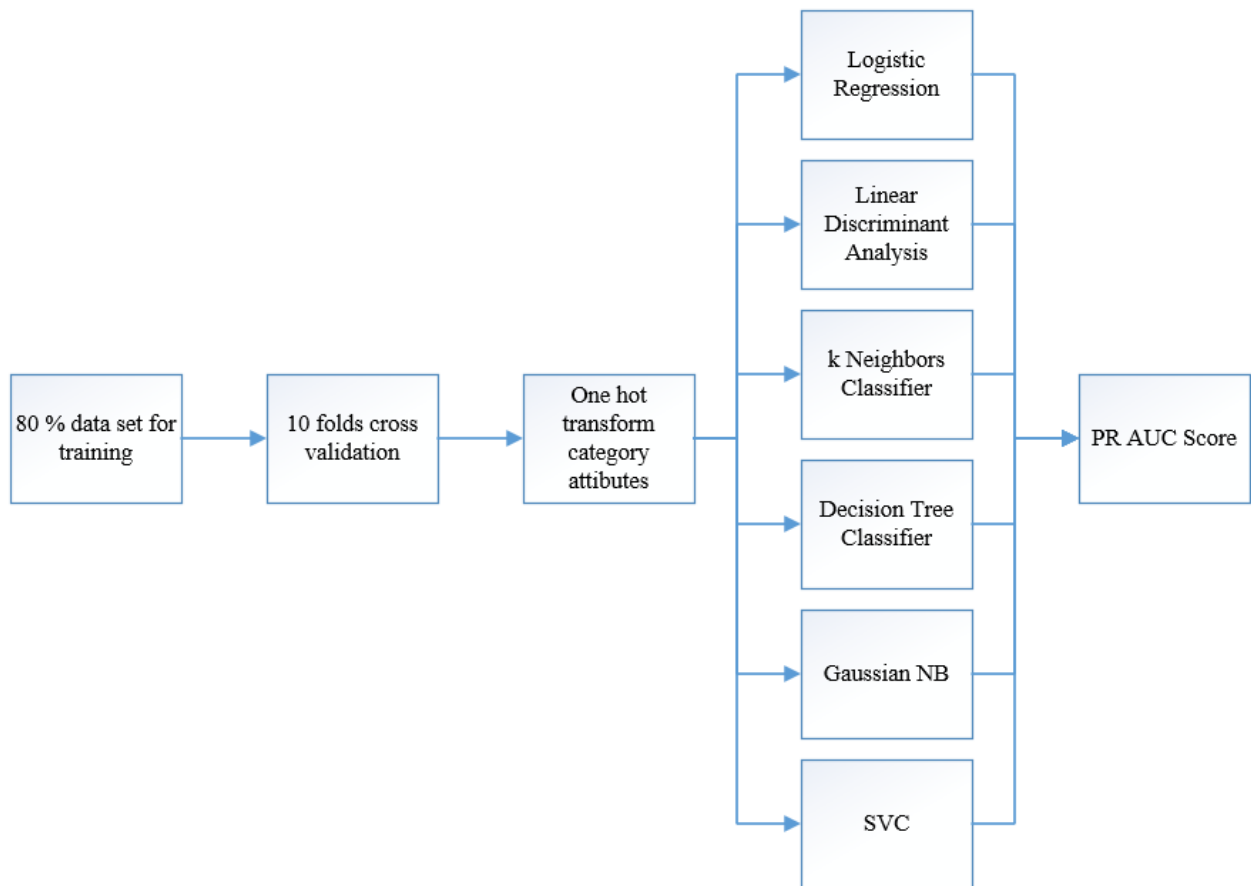


Figure 35: Evaluation baseline workflow.

Figure 35 is a baseline workflow for evaluation. 80 % of the data set will split into ten folds. Each fold saves class distribution the same as the distribution of the original data set. Only category attributes were transformed using a one-hot encoder. A sequence of algorithms will be evaluated;

each model was trained on nine folds and predicted on remain fold. The PR AUC score of each algorithm is an average score per all cross-validation.

Models	Precision-Recall AUC Score (Std)
Logistic Regression	0.881988 (0.007399)
Linear Discriminant Analysis	0.875218 (0.007809)
k Neighbors Classifier	0.681097 (0.010216)
Decision Tree Classifier	0.777429 (0.006443)
Gaussian NB	0.634776 (0.001324)
SVC	0.839620 (0.008465)

Table 6: Precision-Recall AUC score baseline.

Table 6 shows the mean score values, and it suggests that both Logistic Regression and LDA may be worth further study.

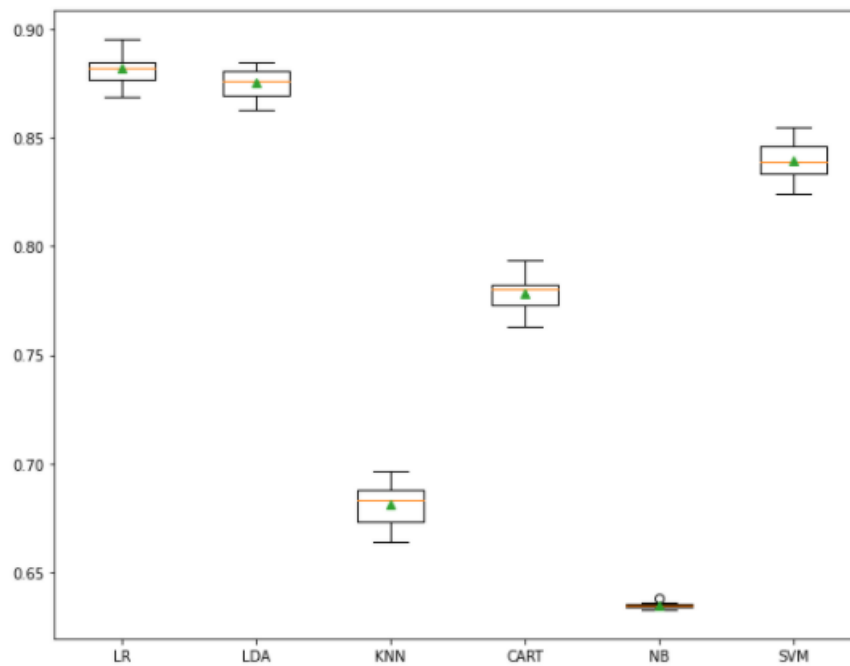


Figure 36: Algorithm comparison baseline.

Figure 36 looks at the distribution of accuracy values calculated across cross-validation folds. It shows graphically using box and whisker plots. The results show a tight distribution for NB, which

is encouraging, suggesting low variance. The poor results for SVM compare with LR and LDA are surprising. The varied distribution of the attributes may be affecting the accuracy of algorithms such as NB.

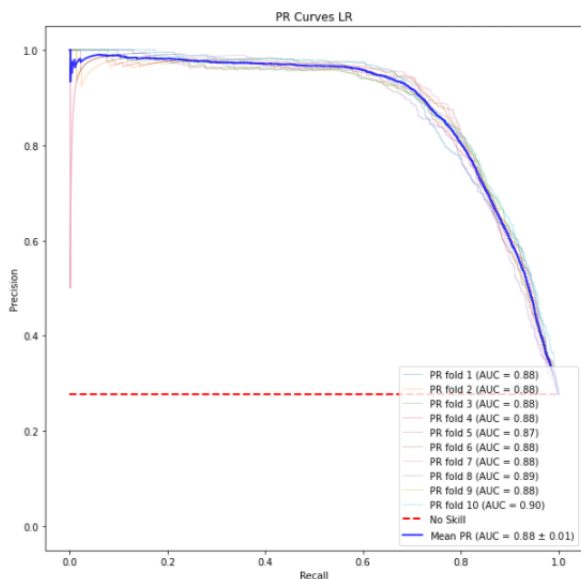


Figure 37: Logistic Regression Cross-Validation.

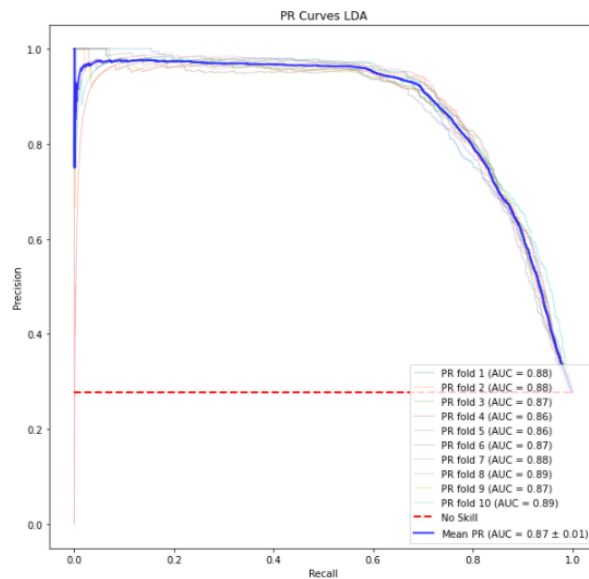


Figure 38: Linear Discriminant Analysis Cross-Validation.

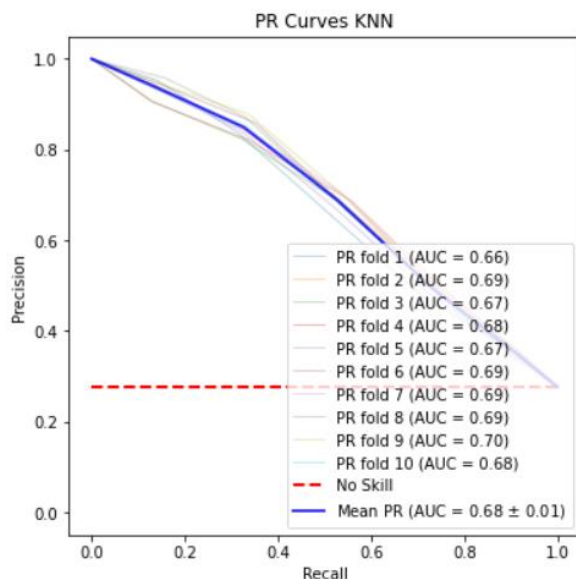


Figure 39: k Neighbors Cross-Validation.

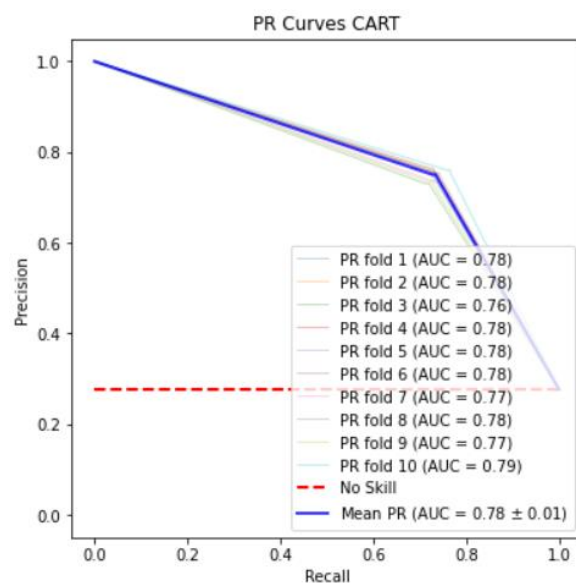


Figure 40: Decision Tree Cross-Validation.

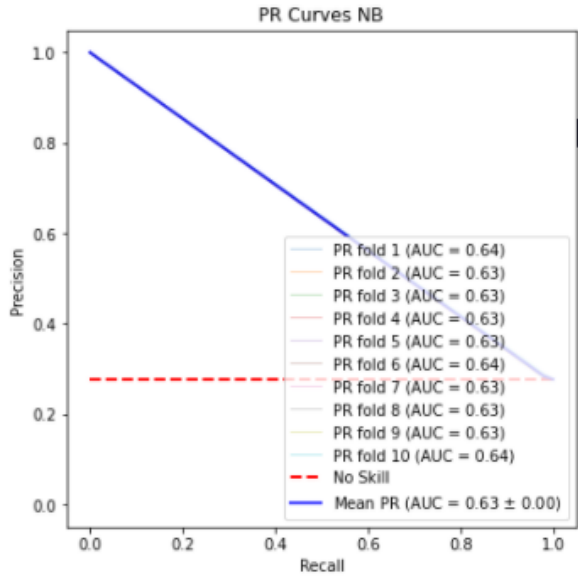


Figure 41: Naive Bayes Cross-Validation.

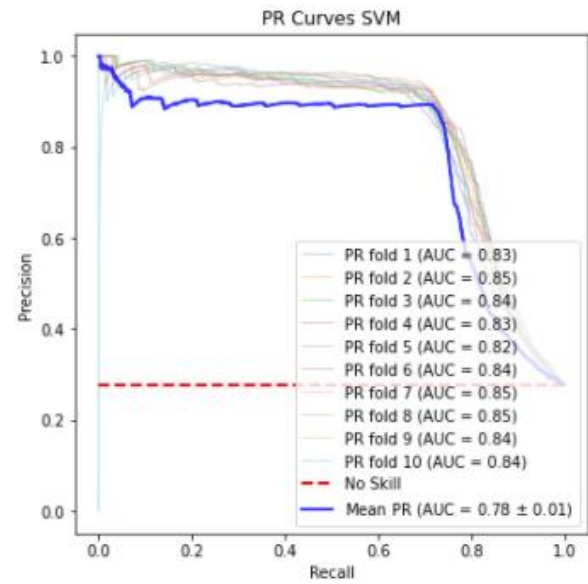


Figure 42: Support Vector Machine Cross-Validation.

Each figure from 37 to 42 plots the Precision-Recall Curve per cross-validation on each evaluated algorithms. The red dot line is the no-skill baseline model, value of the y-axis is the fraction of the number of minority classes per total sample (0.27 in this research data set). The blue curve shows the mean score corresponding to each threshold. Curves are relatively adjacent, which shows a slight difference between folds cross-validation caused by the same class distribution maintained in each fold split.

3.4.1 Evaluate Algorithms: Standardize Data

This section of this study is total the same as Evaluate Algorithms-Baseline above, except all number attributes are scale before modeling. The differing distributions of the raw data may be negatively impacting the skill of some of the algorithms. Let us evaluate the same algorithms with a standardized copy of the dataset. Evaluate Algorithms Standardize Data is where the data transformed such that each attribute has a mean value of zero and a standard deviation of one or the same scaler. It helps seeing how algorithms improve when all number attributes are the same scaler.

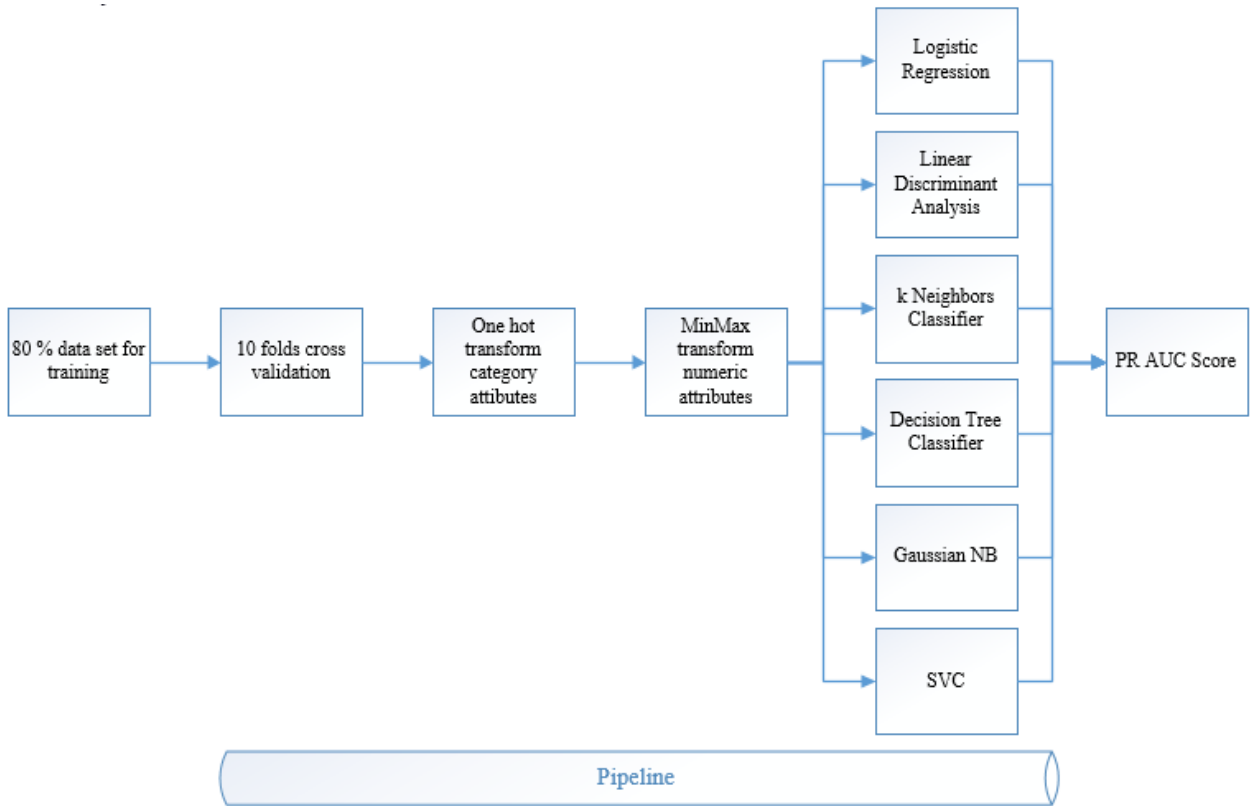


Figure 43: Evaluation algorithms with standardized data workflow.

Models	Precision-Recall AUC Score (Std)	
	Baseline	Standardized Data
Logistic Regression	0.881988 (0.007399)	0.923708 (0.004893)
Linear Discriminant Analysis	0.875218 (0.007809)	0.914196 (0.006199)
k Neighbors Classifier	0.681097 (0.010216)	0.748866 (0.015157)
Decision Tree Classifier	0.777429 (0.006443)	0.854862 (0.006187)
Gaussian NB	0.634776 (0.001324)	0.634795 (0.001276)
SVC	0.839620 (0.008465)	0.916982 (0.004754)

Table 7: Precision-Recall AUC score with standardized data.

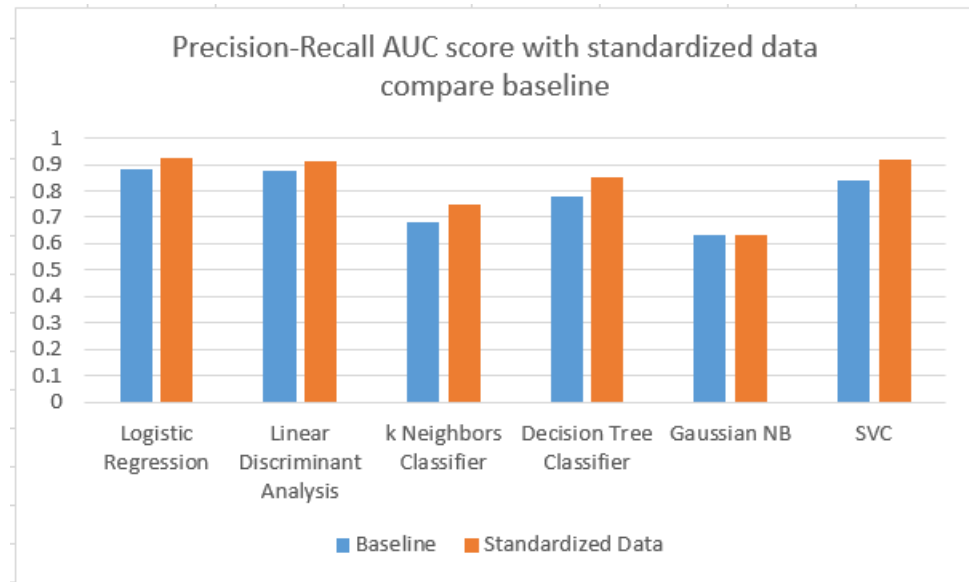


Figure 44: Precision-Recall AUC score with standardized data compare baseline

Table 7 results are listed above. Both Logistic Regression, Linear Discriminant Analysis, and Support Vector Machine are still doing well, even better than before. The results suggest digging deeper into the LR, LDA and SVM algorithms. Configuration beyond the default may likely yield even more accurate models.

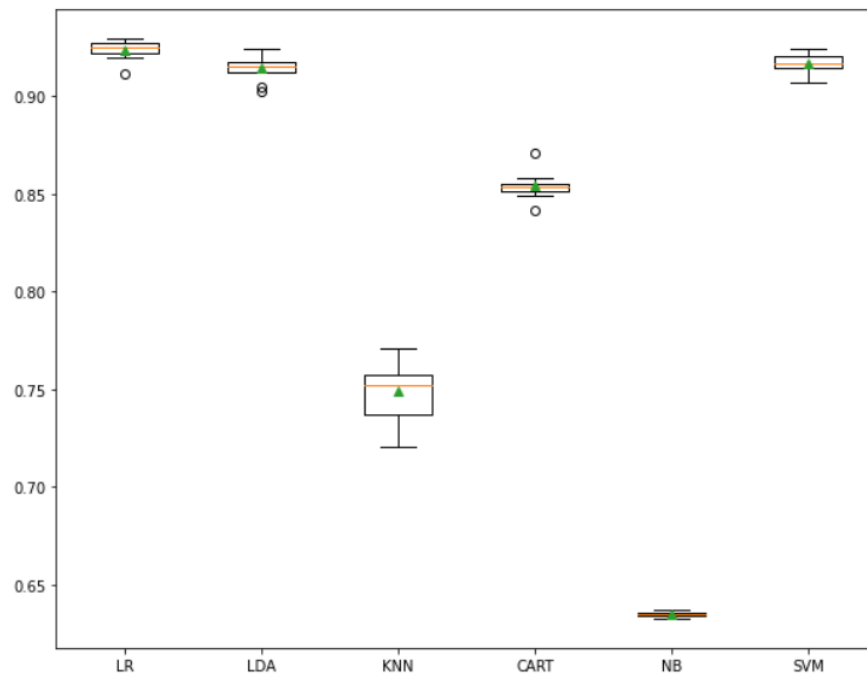


Figure 45: Algorithm comparison with standardize data.

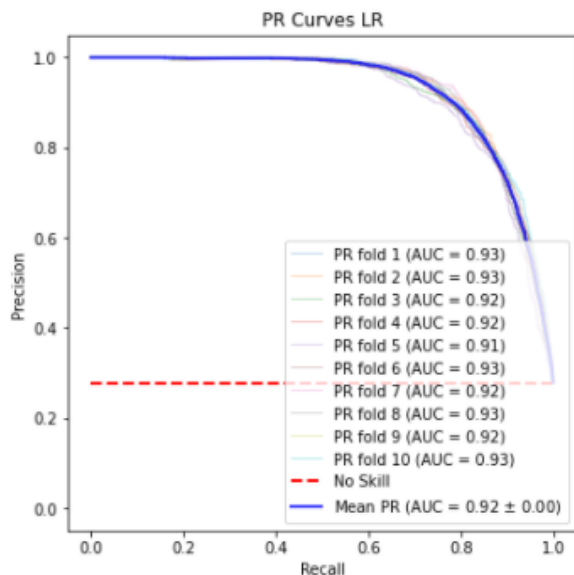


Figure 46: Logistic Regression Cross-Validation with standardized data.

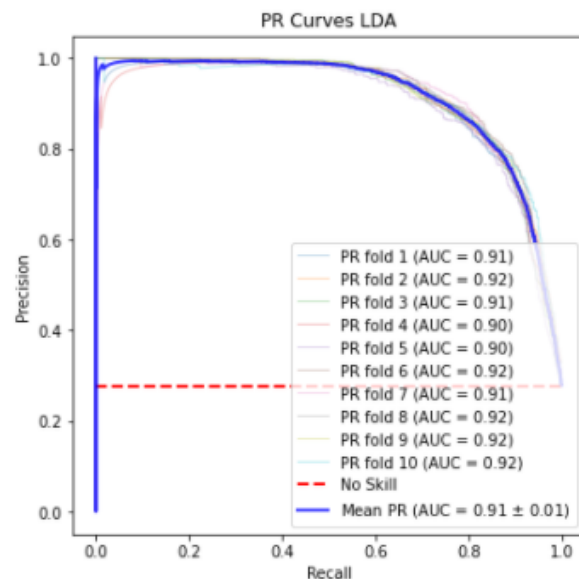


Figure 47: Linear Discriminant Analysis Cross-Validation with standardized data.

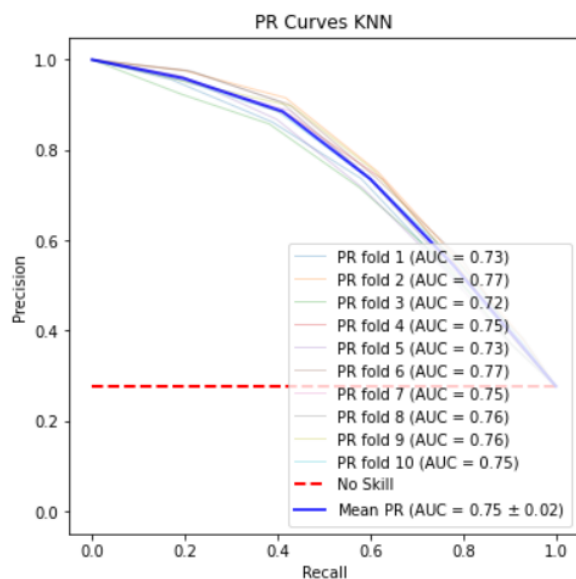


Figure 48: k Neighbors Cross-Validation with standardized data.

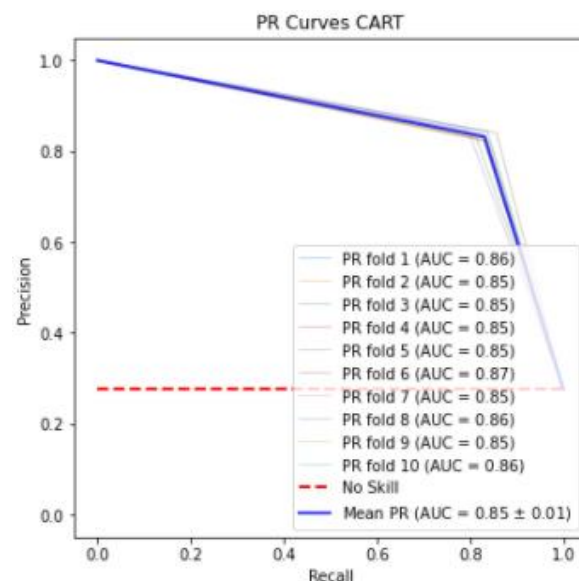


Figure 49: Decision Tree Cross-Validation with standardized data.

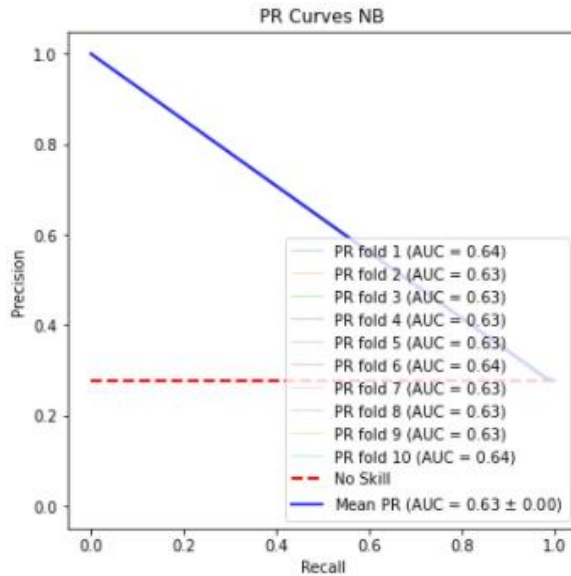


Figure 50: Naive Bayes Cross-Validation with standardized data.

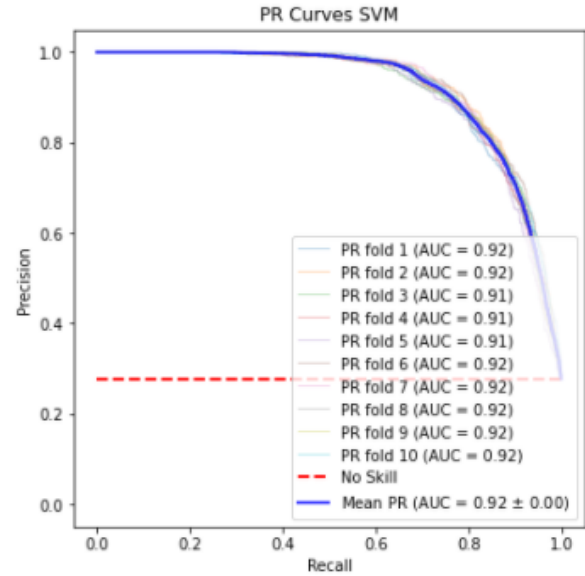


Figure 51: Support Vector Machine Cross-Validation with standardized data.

Figures 45 to 51 point out that almost all algorithms are improved if all numeric attributes are in the same scaler. Compare with the baseline, except the NB algorithm, all remain algorithms increase more than 10% performance.

3.4.2 Improve Results: Algorithm Tuning

This section of the study investigates tuning the parameters for the Logistic Regression algorithm that show promise from the spot-checking in the previous section. The algorithm can start by tuning three parameters are Solver, Penalty, and C values. The 10-fold cross-validation on the standardized training dataset was used for turning.

```

Best: 0.877440 using {'C': 100, 'penalty': 'l2', 'solver': 'newton-cg'}
0.877440 (0.006770) with: {'C': 100, 'penalty': 'l2', 'solver': 'newton-cg'}
0.876230 (0.006852) with: {'C': 100, 'penalty': 'l2', 'solver': 'lbfgs'}
0.877390 (0.006797) with: {'C': 100, 'penalty': 'l2', 'solver': 'liblinear'}
0.876635 (0.006771) with: {'C': 10, 'penalty': 'l2', 'solver': 'newton-cg'}
0.876210 (0.007250) with: {'C': 10, 'penalty': 'l2', 'solver': 'lbfgs'}
0.876665 (0.006801) with: {'C': 10, 'penalty': 'l2', 'solver': 'liblinear'}
0.871355 (0.007534) with: {'C': 1.0, 'penalty': 'l2', 'solver': 'newton-cg'}
0.871290 (0.007476) with: {'C': 1.0, 'penalty': 'l2', 'solver': 'lbfgs'}
0.871396 (0.007486) with: {'C': 1.0, 'penalty': 'l2', 'solver': 'liblinear'}
0.859313 (0.007166) with: {'C': 0.1, 'penalty': 'l2', 'solver': 'newton-cg'}
0.859313 (0.007166) with: {'C': 0.1, 'penalty': 'l2', 'solver': 'lbfgs'}
0.859208 (0.007321) with: {'C': 0.1, 'penalty': 'l2', 'solver': 'liblinear'}
0.846978 (0.006480) with: {'C': 0.01, 'penalty': 'l2', 'solver': 'newton-cg'}
0.846991 (0.006509) with: {'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}
0.846908 (0.006519) with: {'C': 0.01, 'penalty': 'l2', 'solver': 'liblinear'}

```

Figure 52: Results of Tuning LR on the Scaled Dataset.

Figure 52 prints out the configuration that resulted in the highest accuracy and the accuracy of all values. The result shows that the optimal configuration is $C=100$, Penalty = L2, and Solver = newton-CG, but PR-AUC does not improve.

3.4.3 Improve Results: Ensemble Methods

Another way that we can improve the performance of algorithms on this problem is by using ensemble methods. This section will evaluate four different ensemble machine learning algorithms, two boosting and two bagging methods. Boosting Methods: AdaBoost (AB), Gradient Boosting (GBM), and LGBMClassifier. Bagging Methods: Random Forests (RF) and Extra Trees (ET). The experiment pipeline will use the same approach as before, 10-fold cross-validation. The results say that all ensemble techniques methods provide strong accuracy scores in the low 90s (%) with default configurations. Table 8 below identifies the mean and distribution of PR-AUC across the cross-validation folds.

Models	Precision-Recall AUC Score (Std)
AdaBoostClassifier	0.933243 (0.005047)
GradientBoostingClassifier	0.945025 (0.004562)
RandomForestClassifier	0.916121 (0.006834)
ExtraTreesClassifier	0.902806 (0.006040)

LGBMClassifier	0.957052 (0.004170)
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Table 8: Output of Evaluate Ensemble Algorithms.

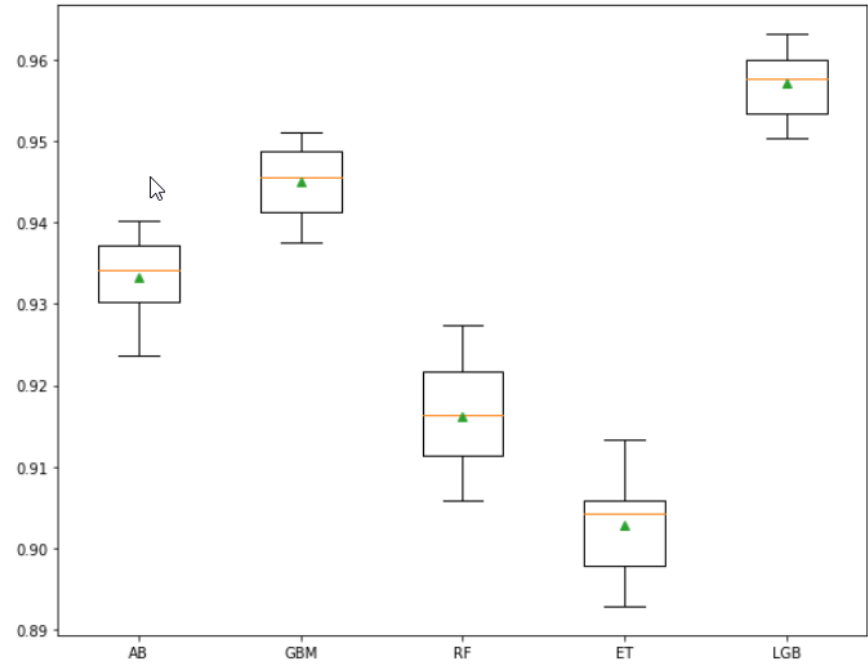


Figure 53: Box and Whisker Plots of Ensemble Performance.

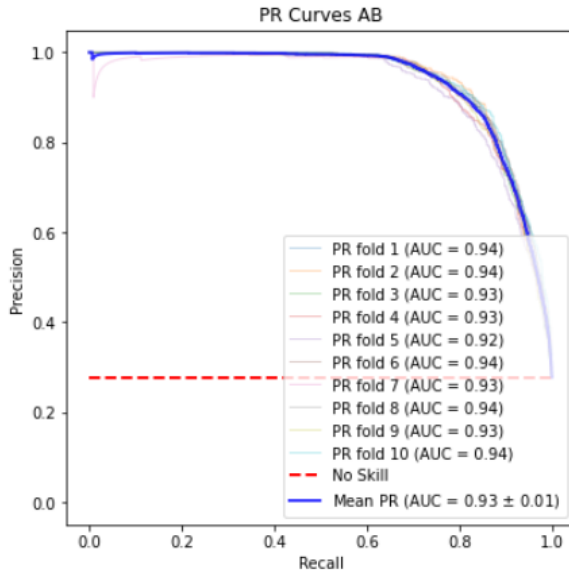


Figure 54: AdaBoost Cross-Validation

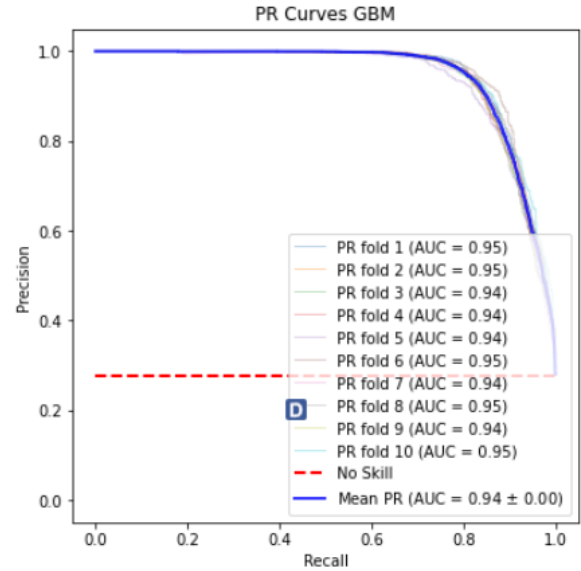


Figure 55: GradientBoosting Cross-Validation.

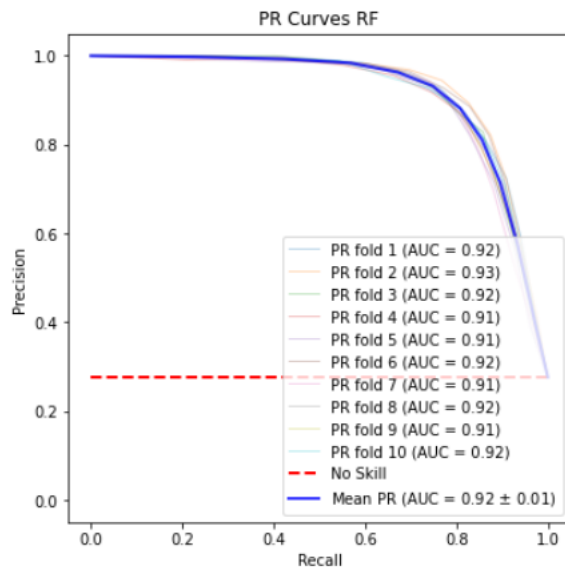


Figure 56: RandomForest Cross-Validation.

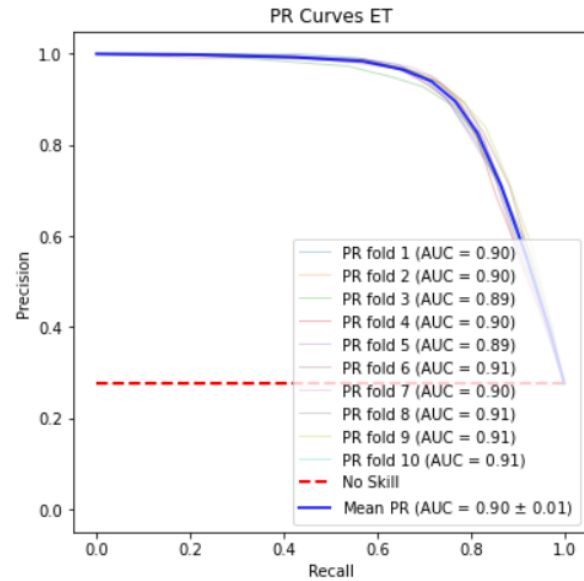


Figure 57: ExtraTrees Cross-Validation.

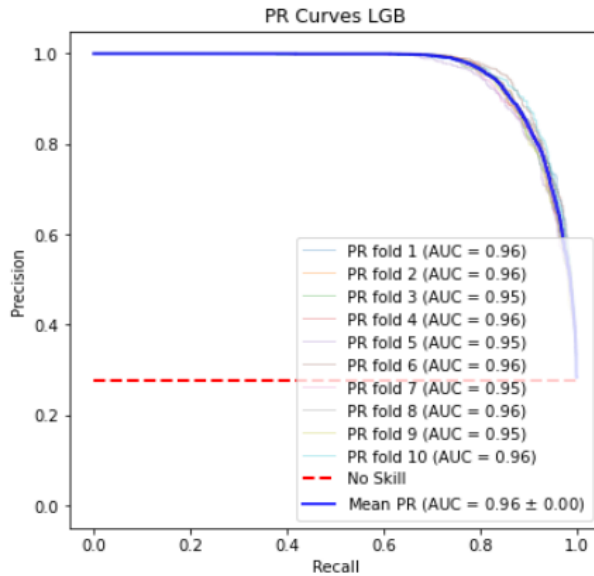


Figure 58: LGBM Cross-Validation.

The results suggest that LGBM is the best model on the data set, reaching a high 0.957 PR-AUC score. Therefore, it is the final model for loan default prediction in this study.

3.4.4 Finalize Model

The LGBM showed the most promise as low complexity and stable model for this problem. In this section, the final model was trained on the entire training dataset and make predictions for the hold-out validation dataset to confirm our findings. A part of the findings was that LGBM performs better when the dataset is standardized so that all attributes have the same scaler.

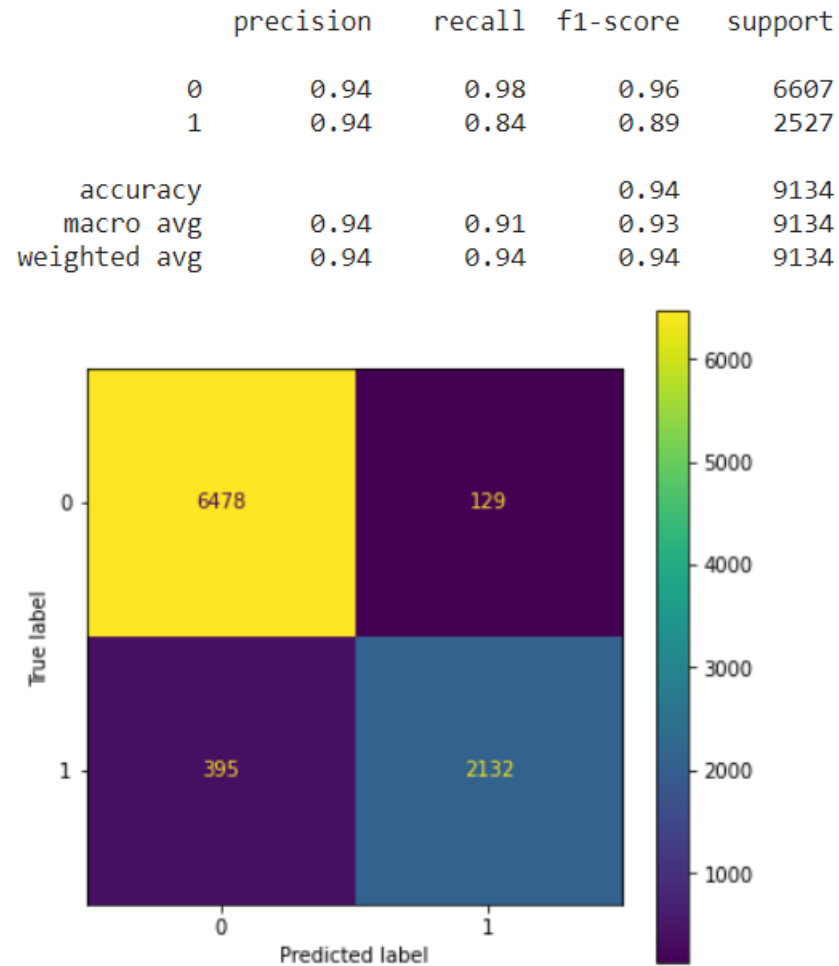


Figure 59: Output of Evaluating LGBM on the Validation Dataset.

Figure 59 shows that the result achieves an accuracy of nearly 94% on the held-out validation dataset. That score matches closely to the expectations of the study. With good predictive results, this research can confidently be used to predict the loan default on VietCredit's loan data set.

Chapter 4

Conclusions and Perspectives

Nowadays, the loan business becomes more and famous, and many people apply for loans for various reasons. However, there are cases where people do not repay the bulk of the loan amount to the bank, which results in substantial financial loss. Hence, if there is a way to efficiently classify the loaners in advance, it would significantly prevent financial loss.

In this study, the loan application data set was collected from five transaction systems of our company VietCredit. After the data set was analyzed and summarized to expose the best structure of the data before modeling, the data set was pre-processed as cleaning and transforming. The strategies to deal with missing values and imbalanced data sets were covered; the category attributes transformed using the one-hot encoder, while the numeric attributes transformed using a min-max scaler. Then algorithm evaluation metrics were investigated base on characteristics of the data set. Precision-recall curve and PR AUC score recommended for highly skewed domains, so it is used in this study to evaluate and compare the performance of each model. After tuning algorithms and improving by ensemble methods, the LGBM ensemble method becomes the best classifier with a high PR-AUC score of 0.957. On the held-out test data set, the mean accuracy of prediction archives 94%, and that matches closely to the expectations of the research. So The LGBM is the best-suited model for predicting loan default problems.

Our future researchers in this field may apply the resampling techniques to the datasets. This technique helps to reduce the imbalance ratio of datasets which in turn produces better classification results. The number of machine learning algorithms is perhaps more than a hundred, so the study will test more algorithms to find out better ones in the future. Some automatic feature selection techniques have not been used so far. Therefore, we also plan to apply Recursive Feature Elimination (RFE) to remove redundancy attributes, using Principal Component Analysis (PCA) to transform the dataset into a compressed form to enhance the performance of models. We will also try to conduct experiments on larger data sets or try to tune the ensemble model to achieve the state-of-art performance of the model.

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